

Pine Island Sound Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

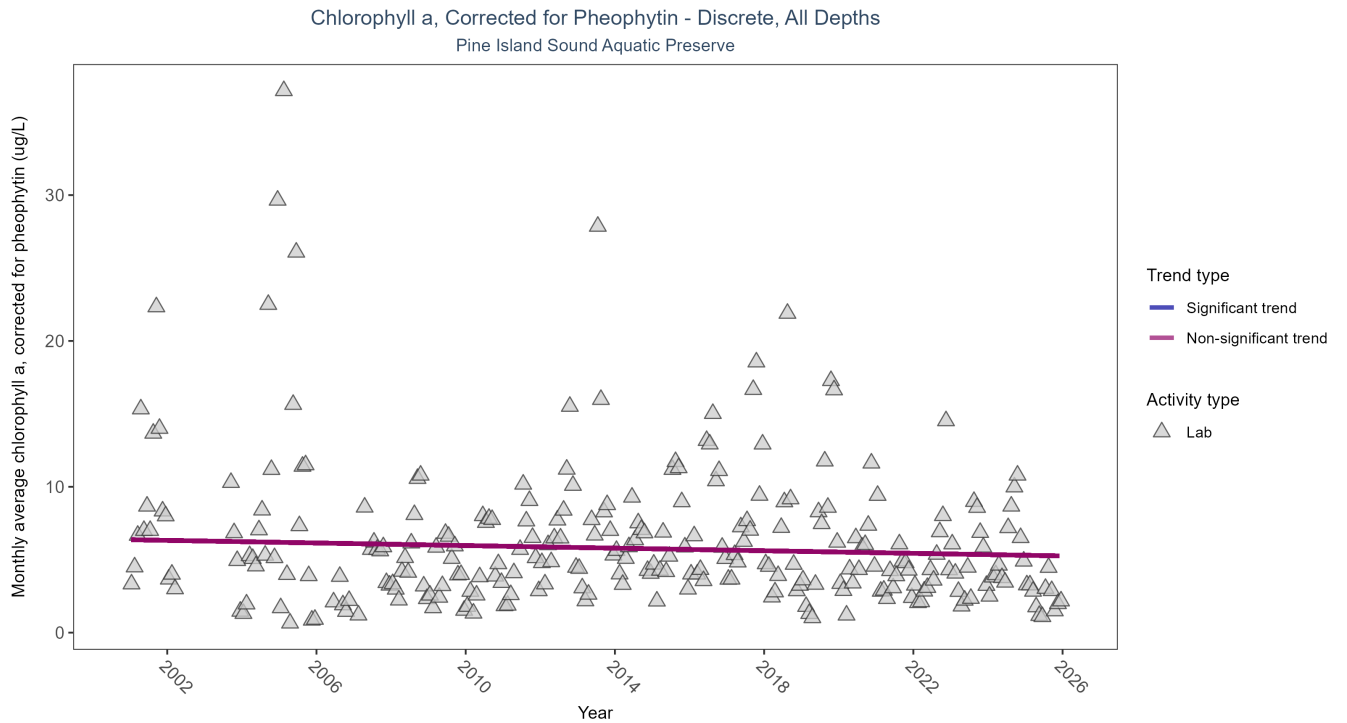


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	4182	25	2001 - 2025	3.9	-0.0863	6.3699	-0.0444	0.0669

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2001 and 2025.

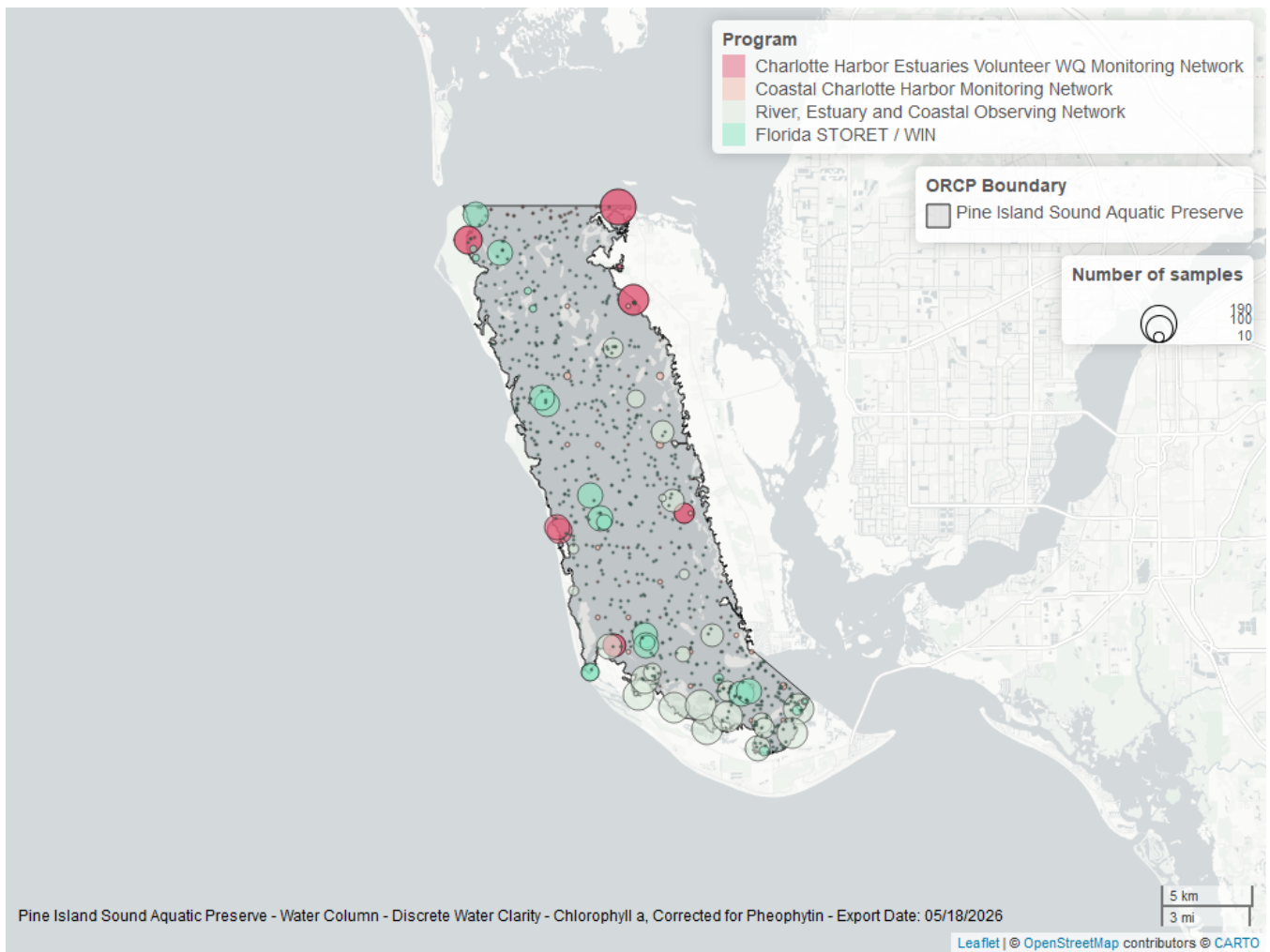


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
303	2073	2012	2025
5002	1276	2001	2025
476	707	2008	2025
513	185	2003	2025

Program names:

303 - River, Estuary and Coastal Observing Network¹

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³

513 - Coastal Charlotte Harbor Monitoring Network⁴

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

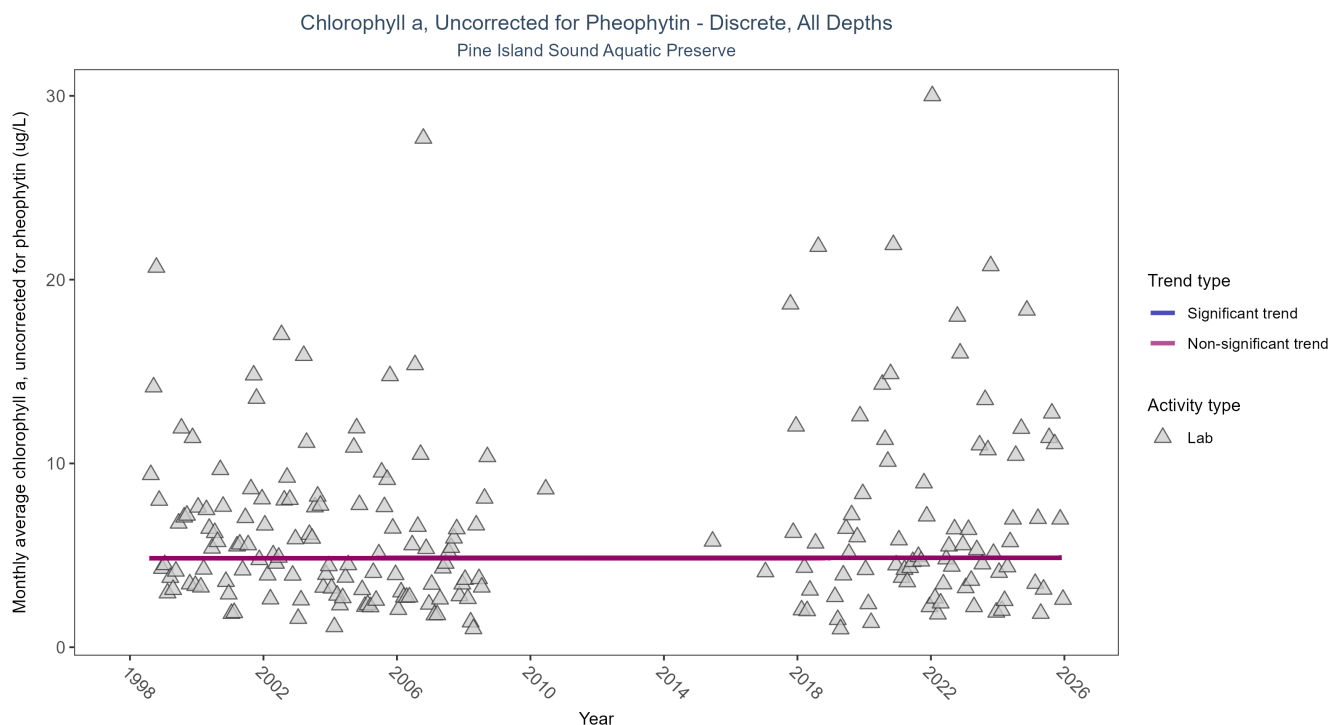


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1270	22	1998 - 2025	4.3674	0.0017	4.8372	0.0011	1

Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1998 and 2025.

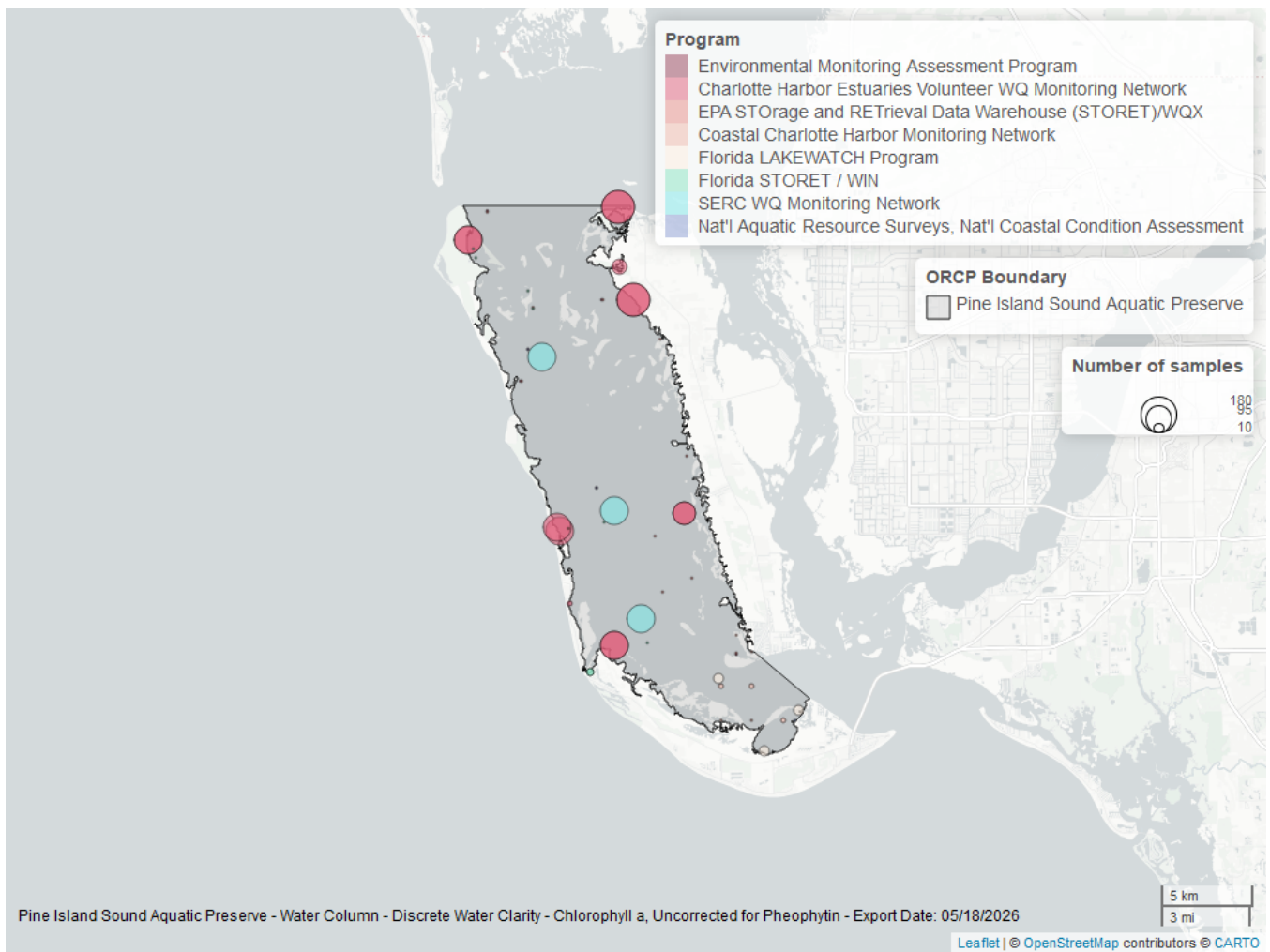


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
476	831	1998	2025
509	348	1999	2008
514	59	2001	2002
5002	22	2005	2021
513	16	2002	2004
103	14	2002	2015
115	4	2002	2004
118	3	2006	2010

Program names:

[476](#) - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³

509 - SERC Water Quality Monitoring Network⁵

514 - Florida LAKEWATCH Program⁶

5002 - Florida STORET / WIN²

513 - Coastal Charlotte Harbor Monitoring Network⁴
 103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX⁷
 115 - Environmental Monitoring Assessment Program⁸
 118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁹

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

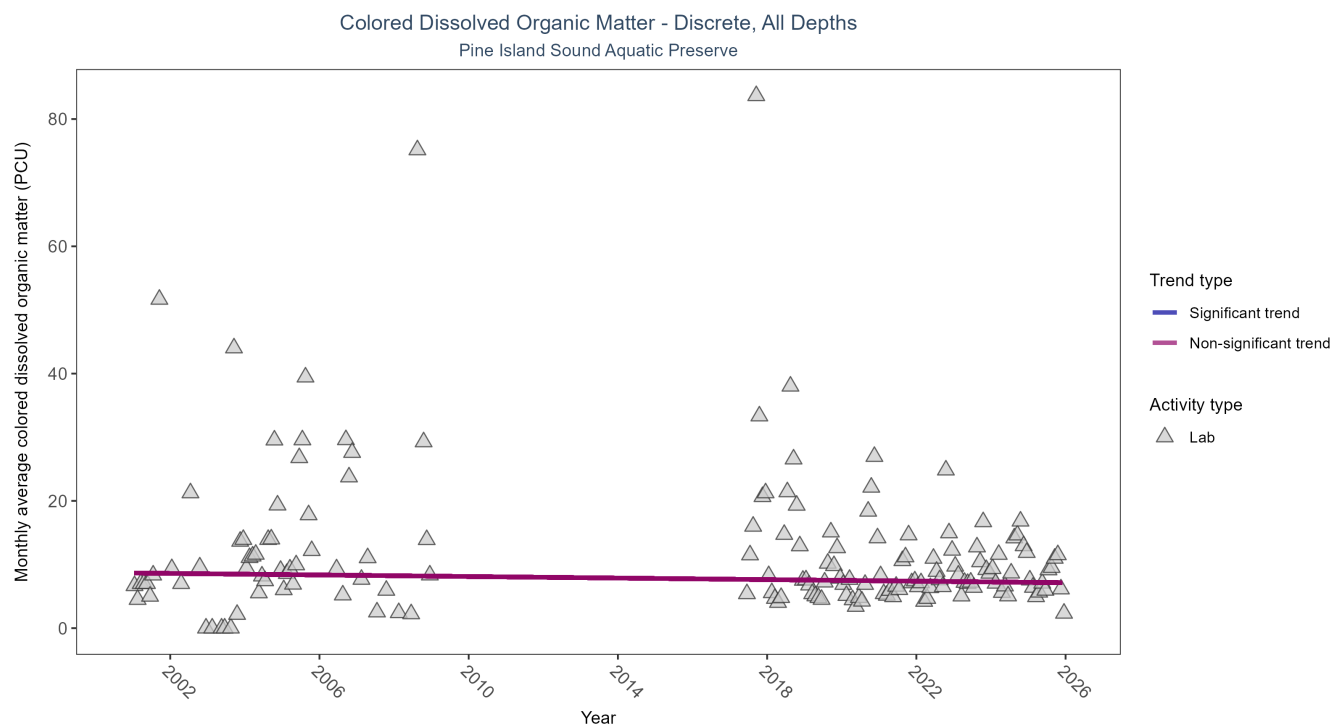


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1706	17	2001 - 2025	7.59	-0.0879	8.6629	-0.0609	0.1634

Colored dissolved organic matter showed no detectable trend between 2001 and 2025.

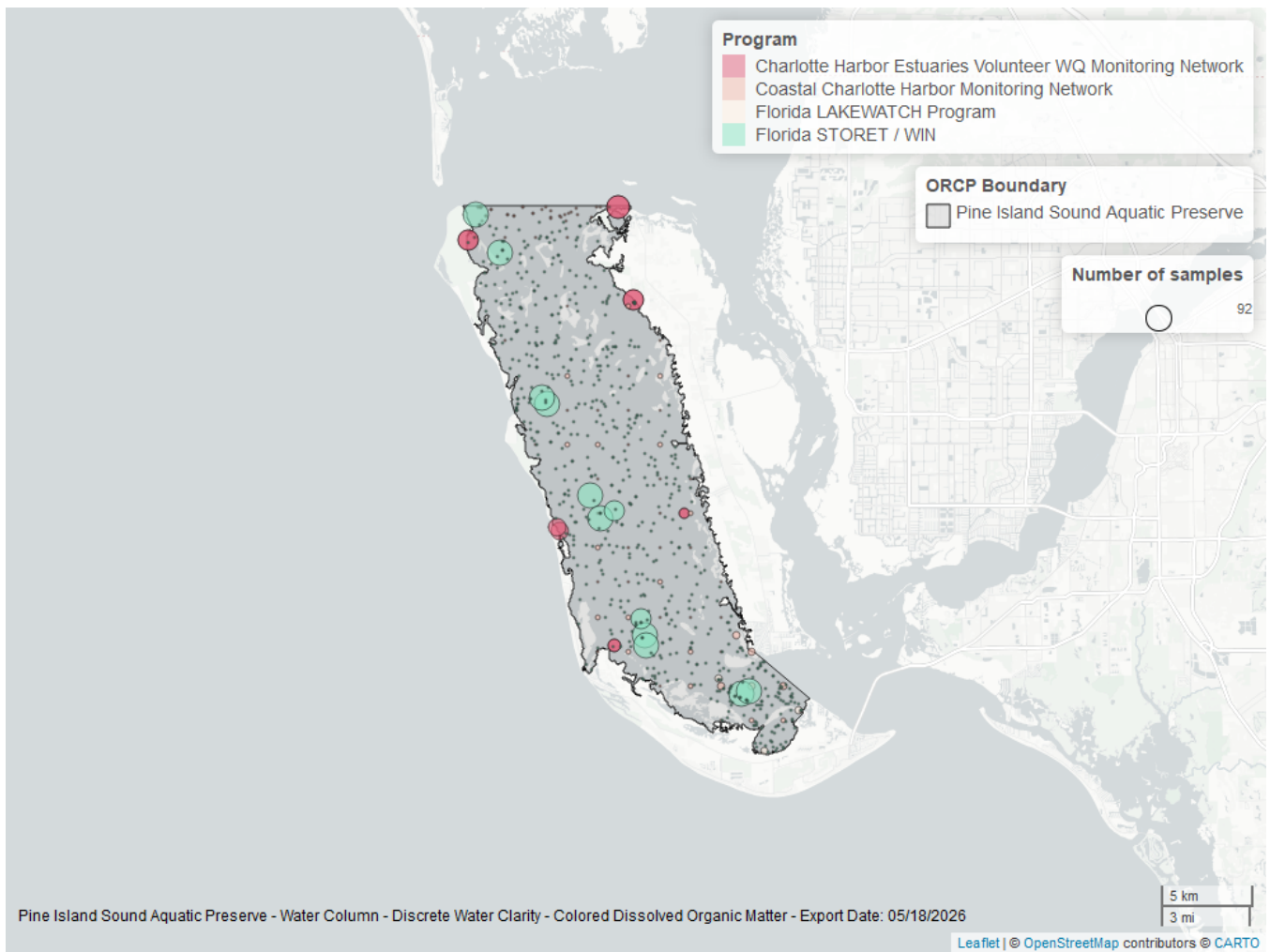


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
5002	1198	2018	2025
476	292	2017	2025
513	182	2002	2025
514	35	2001	2002

Program names:

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³

513 - Coastal Charlotte Harbor Monitoring Network⁴

514 - Florida LAKEWATCH Program⁶

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

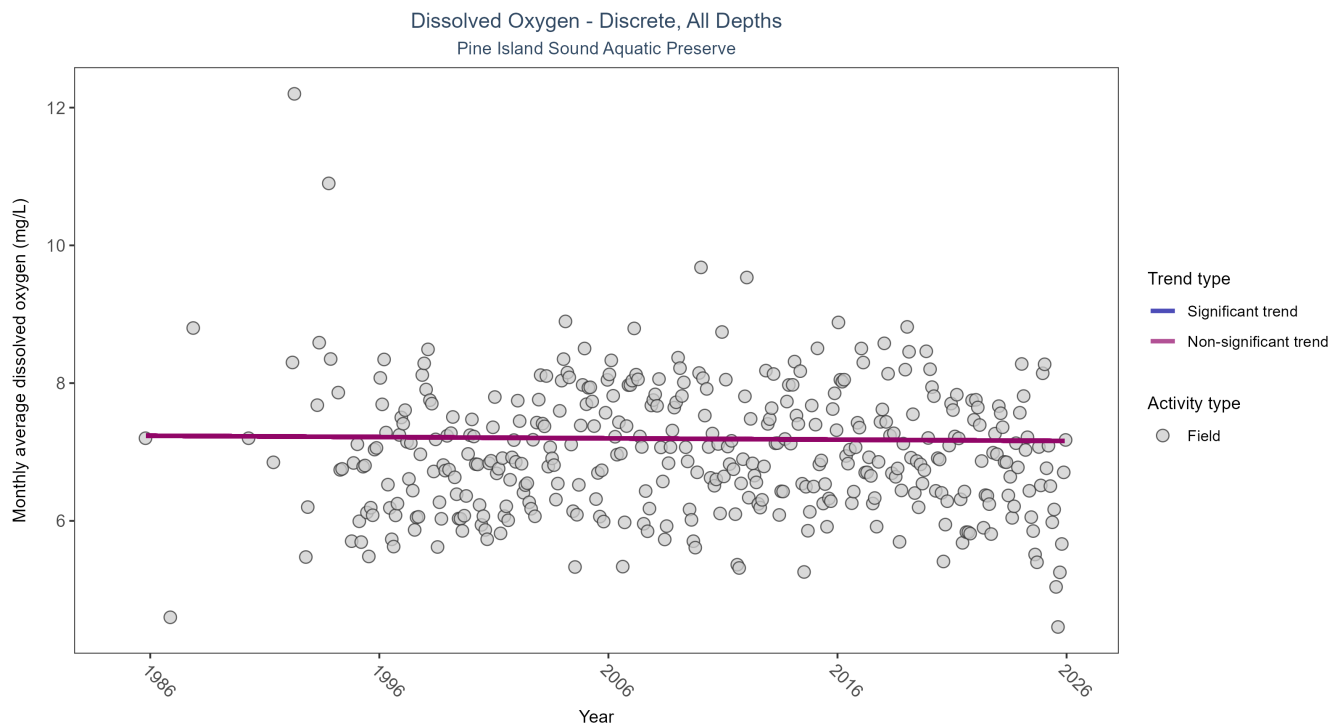


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	36431	39	1985 - 2025	6.9	-0.0193	7.2377	-0.0019	0.5557

Dissolved oxygen showed no detectable trend between 1985 and 2025.

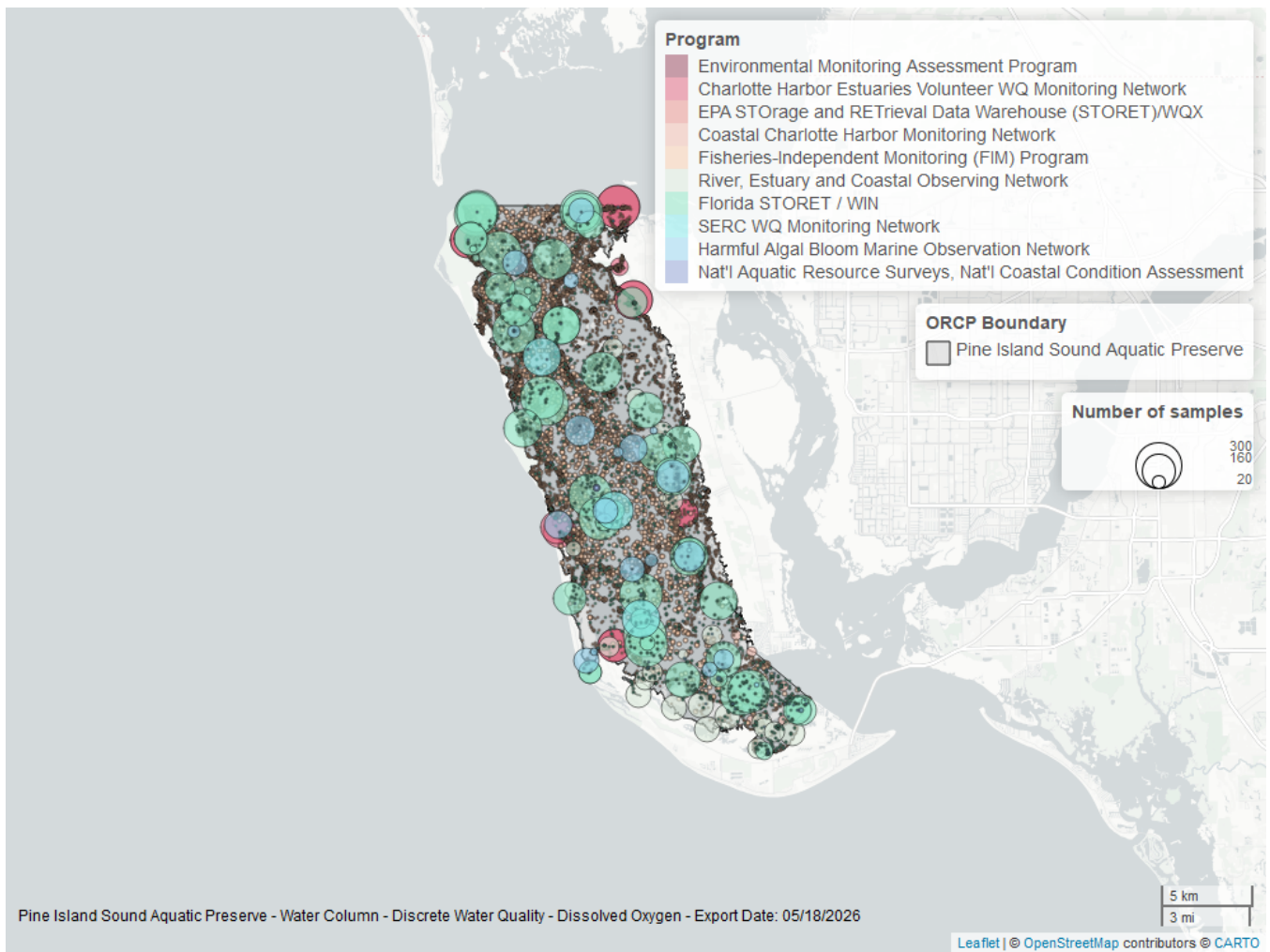


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
69	21437	1990	2024
5002	9822	1987	2025
303	1598	2012	2025
95	1530	1985	2018
476	1249	1998	2025
509	696	1999	2008
513	200	2002	2025
118	16	2006	2020
115	14	2002	2004
103	5	2015	2015

Program names:

69 - Fisheries-Independent Monitoring (FIM) Program¹⁰

5002 - Florida STORET / WIN²

303 - River, Estuary and Coastal Observing Network¹
95 - Harmful Algal Bloom Marine Observation Network¹¹
476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³
509 - SERC Water Quality Monitoring Network⁵
513 - Coastal Charlotte Harbor Monitoring Network⁴
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁹
115 - Environmental Monitoring Assessment Program⁸
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

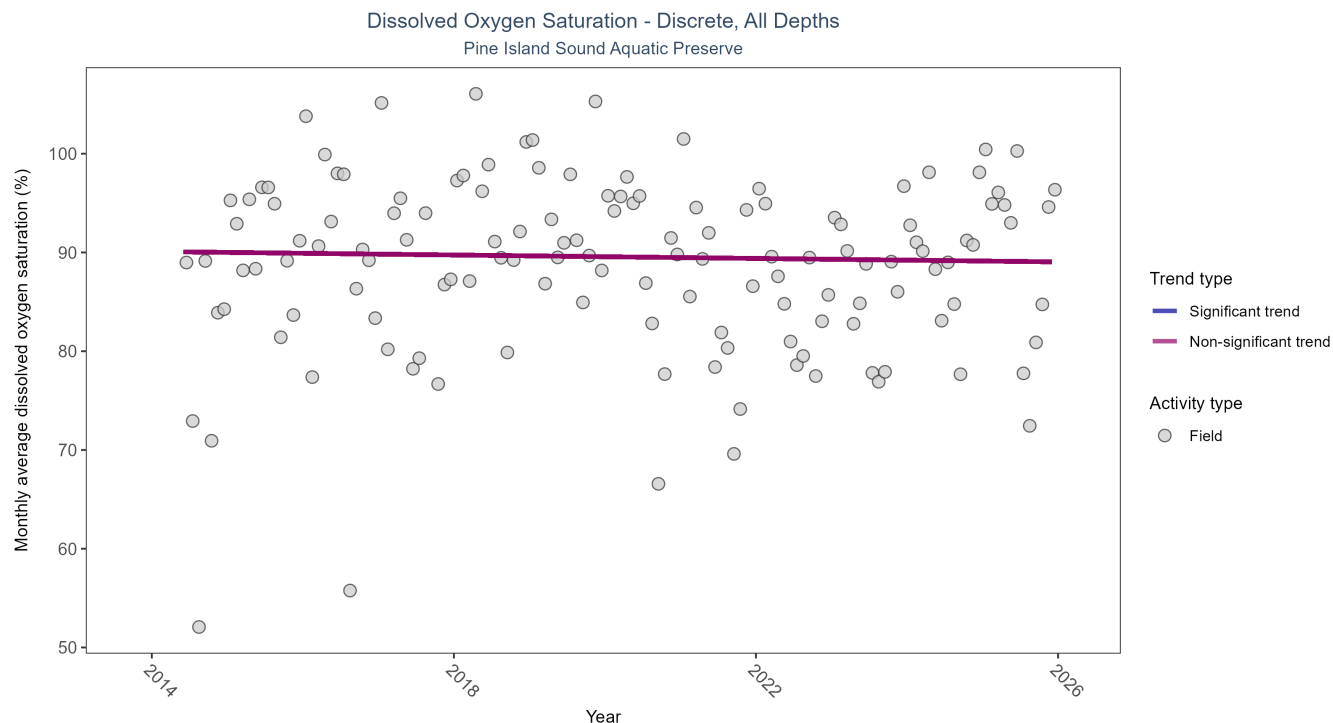


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	3650	12	2014 - 2025	91.0886	-0.0445	90.0917	-0.0873	0.5424

Dissolved oxygen saturation showed no detectable trend between 2014 and 2025.

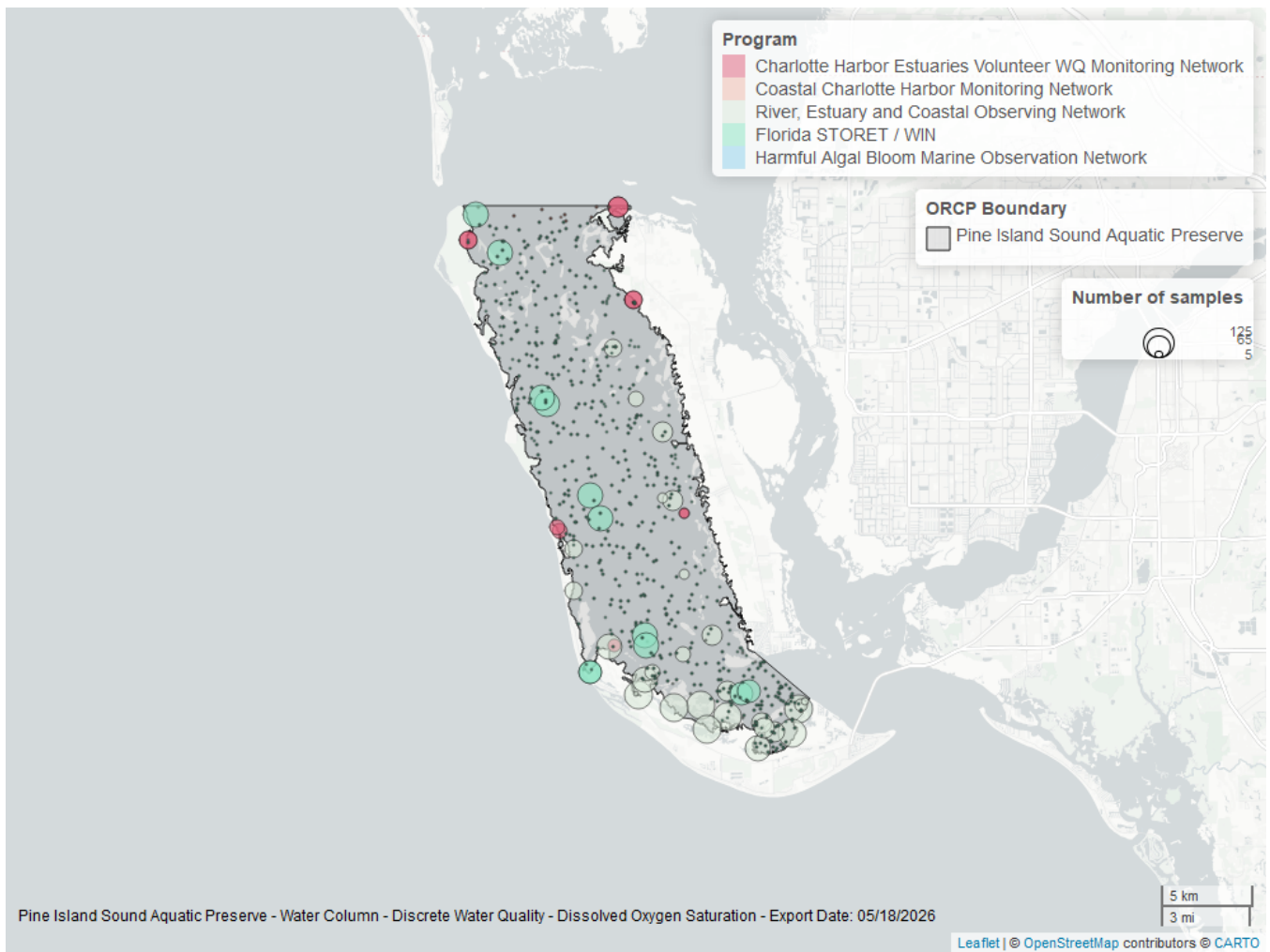


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
303	1868	2014	2025
5002	1552	2014	2025
476	230	2017	2025
513	19	2023	2025
95	2	2016	2016

Program names:

303 - River, Estuary and Coastal Observing Network¹

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³

513 - Coastal Charlotte Harbor Monitoring Network⁴

95 - Harmful Algal Bloom Marine Observation Network¹¹

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

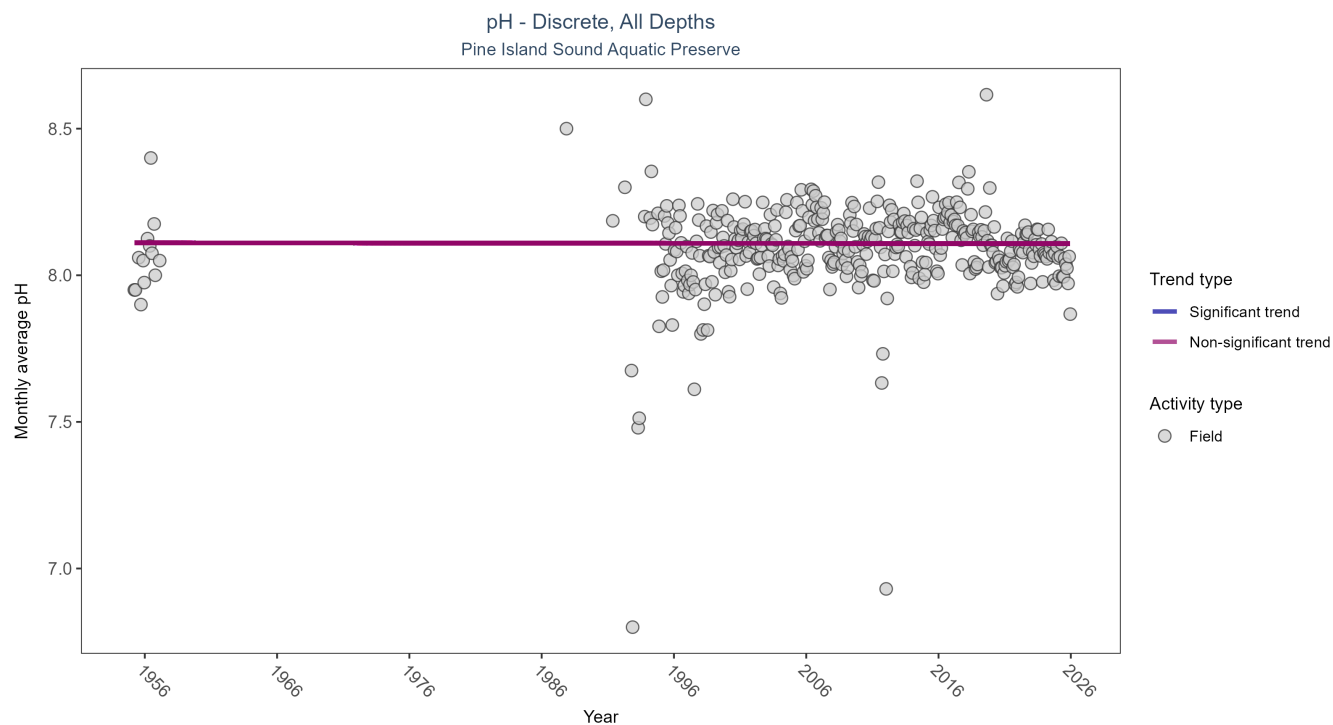


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	34027	39	1955 - 2025	8.1	-0.0039	8.1107	0	0.9011

pH showed no detectable trend between 1955 and 2025.

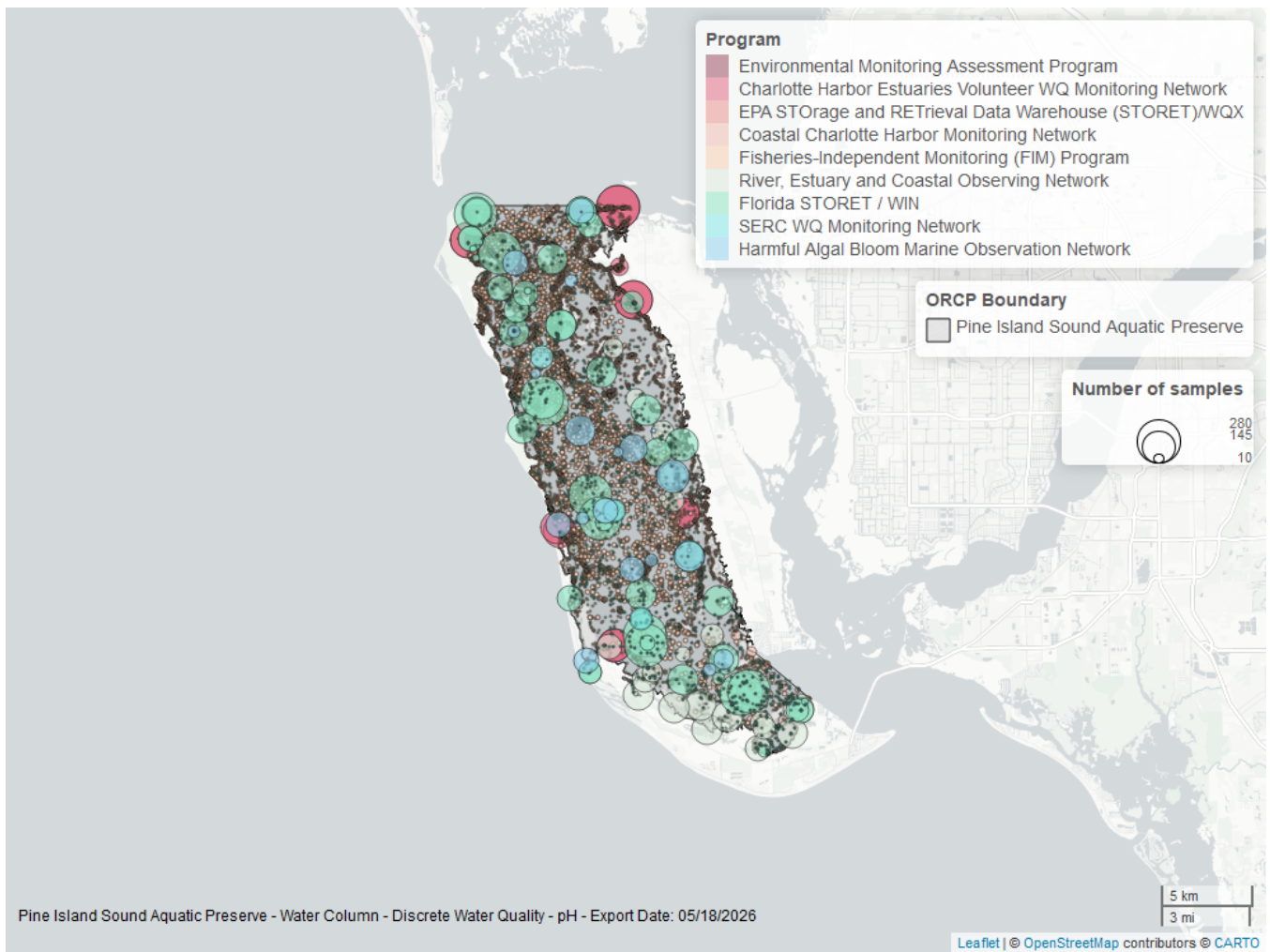


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
69	21220	1991	2024
5002	7607	1987	2025
303	2067	2012	2025
95	1464	1955	2018
476	1251	1998	2025
509	270	2001	2008
513	201	2002	2025
115	14	2002	2004
103	5	2015	2015

Program names:

69 - Fisheries-Independent Monitoring (FIM) Program¹⁰

5002 - Florida STORET / WIN²

303 - River, Estuary and Coastal Observing Network¹

95 - Harmful Algal Bloom Marine Observation Network¹¹
476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³
509 - SERC Water Quality Monitoring Network⁵
513 - Coastal Charlotte Harbor Monitoring Network⁴
115 - Environmental Monitoring Assessment Program⁸
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

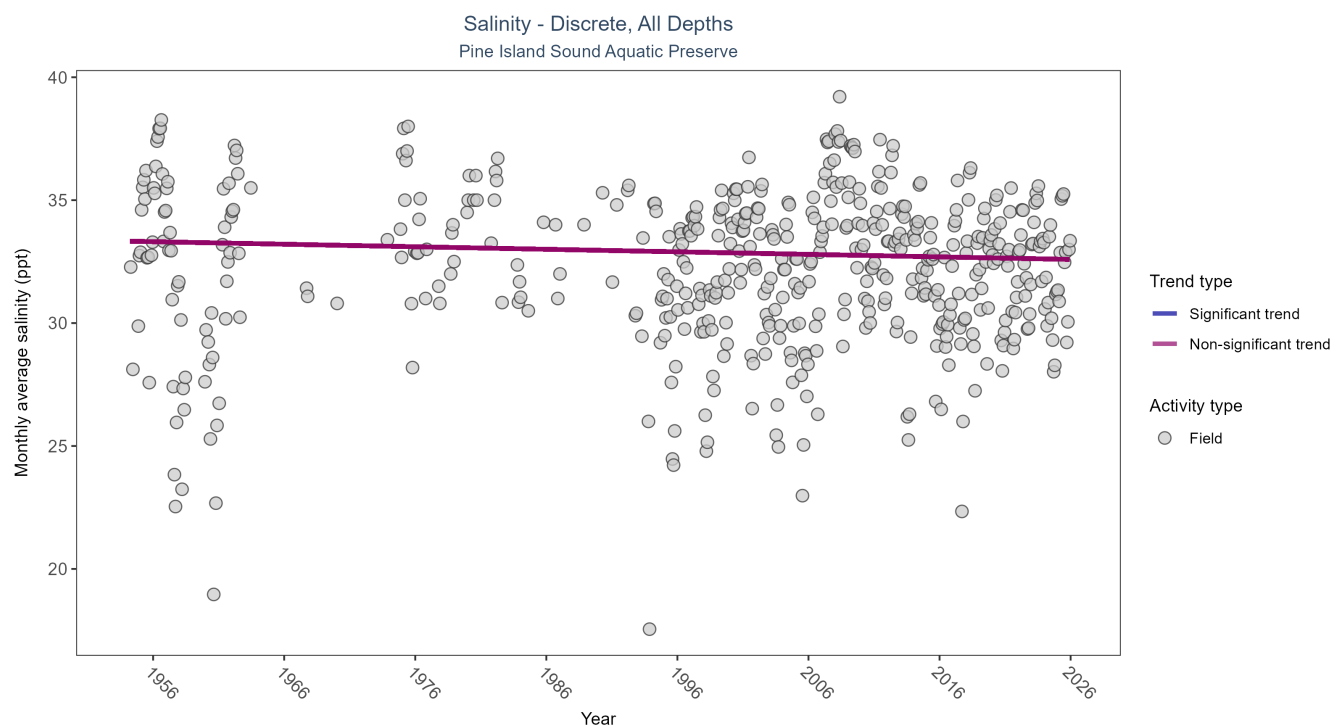


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	35234	64	1954 - 2025	32.8	-0.0526	33.3321	-0.0103	0.0909

Salinity showed no detectable trend between 1954 and 2025.

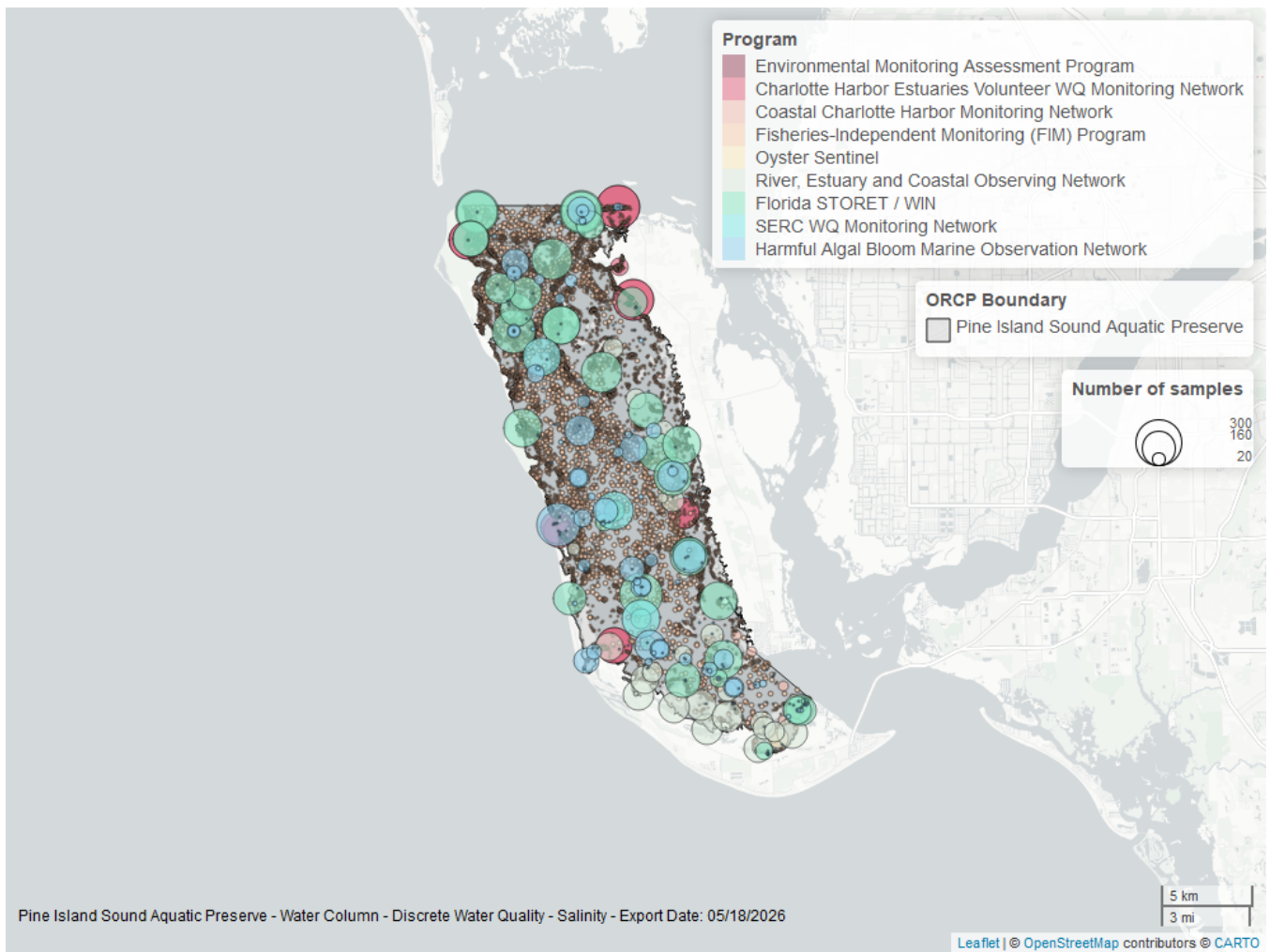


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
69	21553	1990	2024
5002	6231	1995	2025
95	2924	1954	2018
303	2265	2012	2025
476	1310	1998	2025
509	702	1999	2008
513	169	2002	2008
456	71	1970	2012
115	14	2002	2004

Program names:

69 - Fisheries-Independent Monitoring (FIM) Program¹⁰

5002 - Florida STORET / WIN²

95 - Harmful Algal Bloom Marine Observation Network¹¹

303 - River, Estuary and Coastal Observing Network¹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³

509 - SERC Water Quality Monitoring Network⁵

513 - Coastal Charlotte Harbor Monitoring Network⁴

456 - Oyster Sentinel¹²

115 - Environmental Monitoring Assessment Program⁸

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

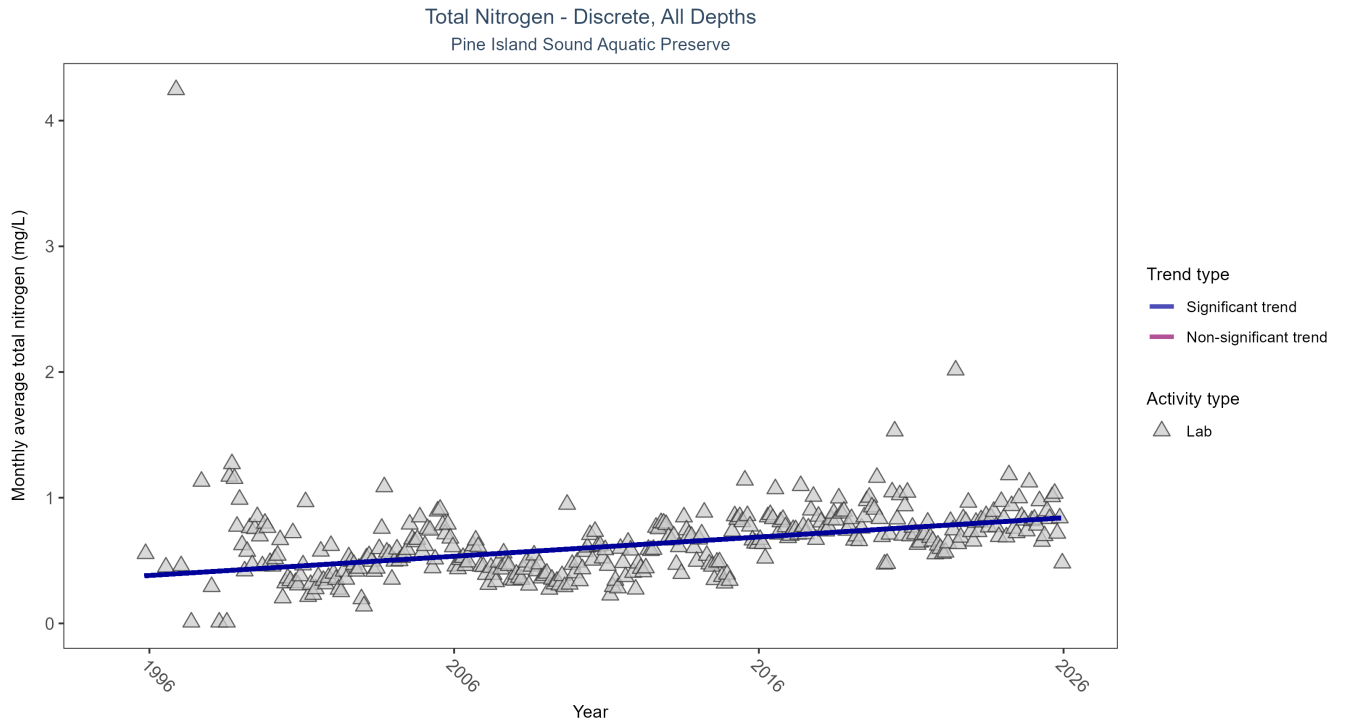


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	6361	31	1995 - 2025	0.62	0.3703	0.3659	0.0153	0

Monthly average total nitrogen increased by 0.01 mg/L per year.

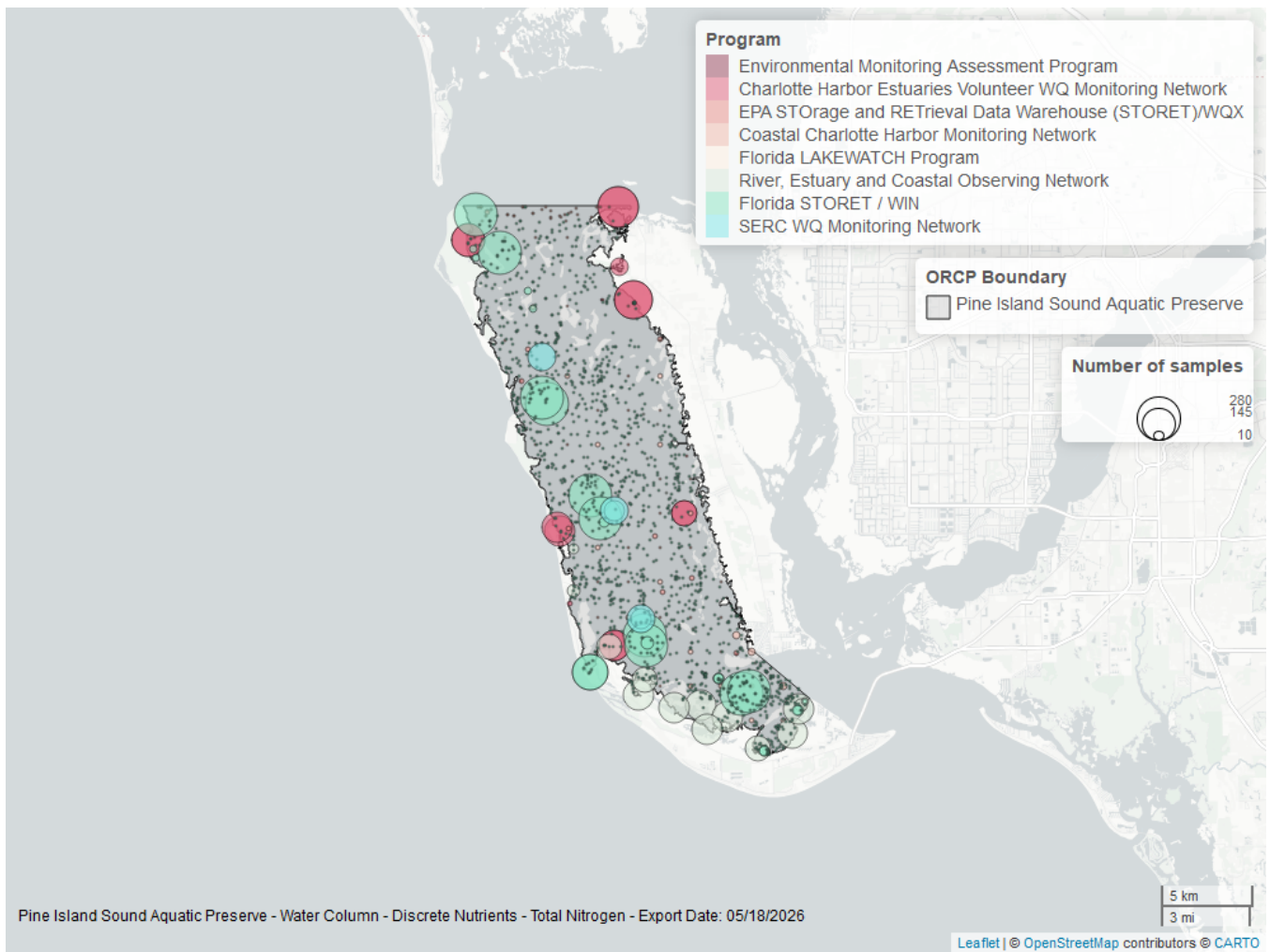


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
5002	3273	1995	2025
303	1378	2012	2025
476	1134	1998	2025
509	351	1999	2008
513	154	2002	2025
514	59	2001	2002
103	37	2002	2015
115	4	2002	2004

Program names:

5002 - Florida STORET / WIN²

303 - River, Estuary and Coastal Observing Network¹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³

509 - SERC Water Quality Monitoring Network⁵

513 - Coastal Charlotte Harbor Monitoring Network⁴
 514 - Florida LAKEWATCH Program⁶
 103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷
 115 - Environmental Monitoring Assessment Program⁸

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

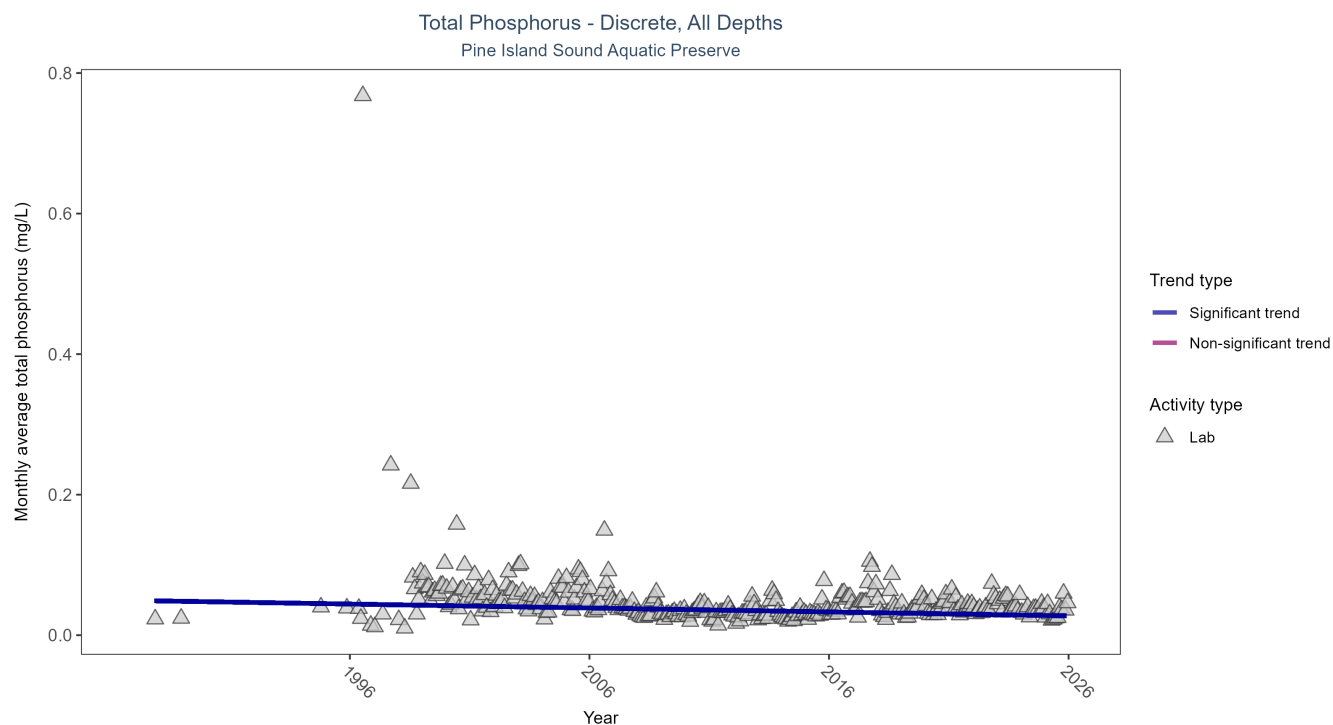


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	6193	34	1987 - 2025	0.036	-0.1869	0.0493	-0.0006	0

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

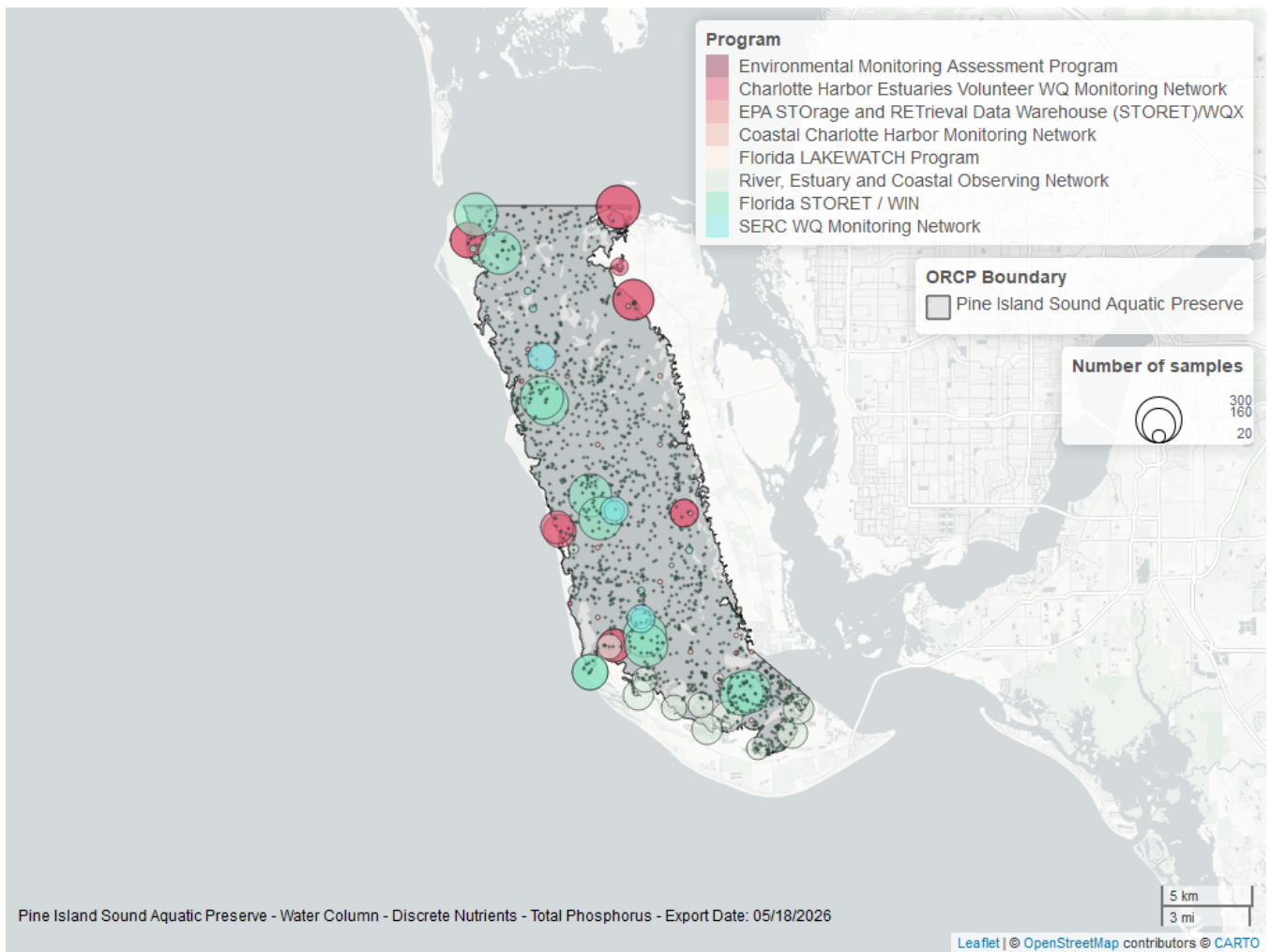


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5002	3158	1987	2025
476	1273	1998	2025
303	1242	2012	2025
509	348	1999	2008
513	165	2003	2025
514	59	2001	2002
103	27	2002	2015
115	4	2002	2004

Program names:

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³

303 - River, Estuary and Coastal Observing Network¹

509 - SERC Water Quality Monitoring Network⁵

513 - Coastal Charlotte Harbor Monitoring Network⁴
514 - Florida LAKEWATCH Program⁶
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷
115 - Environmental Monitoring Assessment Program⁸

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

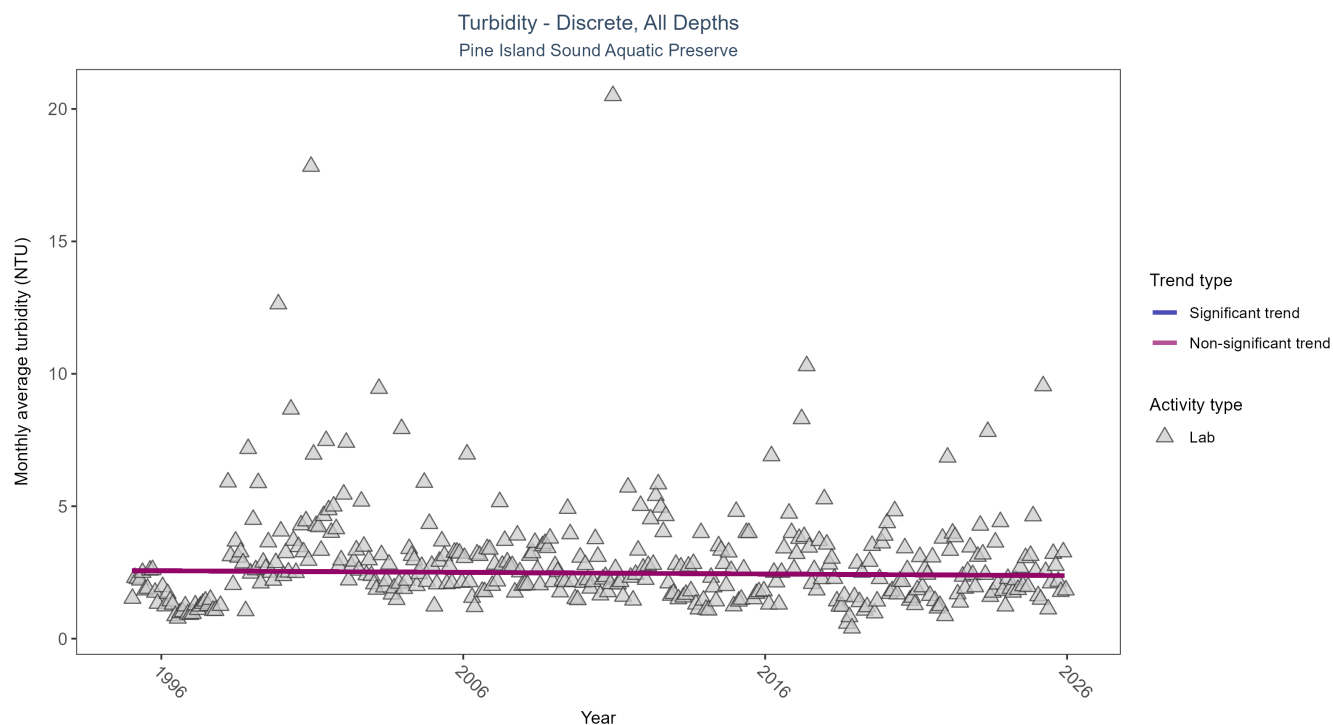


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	6844	31	1995 - 2025	2	-0.0257	2.5692	-0.0061	0.4926

Turbidity showed no detectable trend between 1995 and 2025.

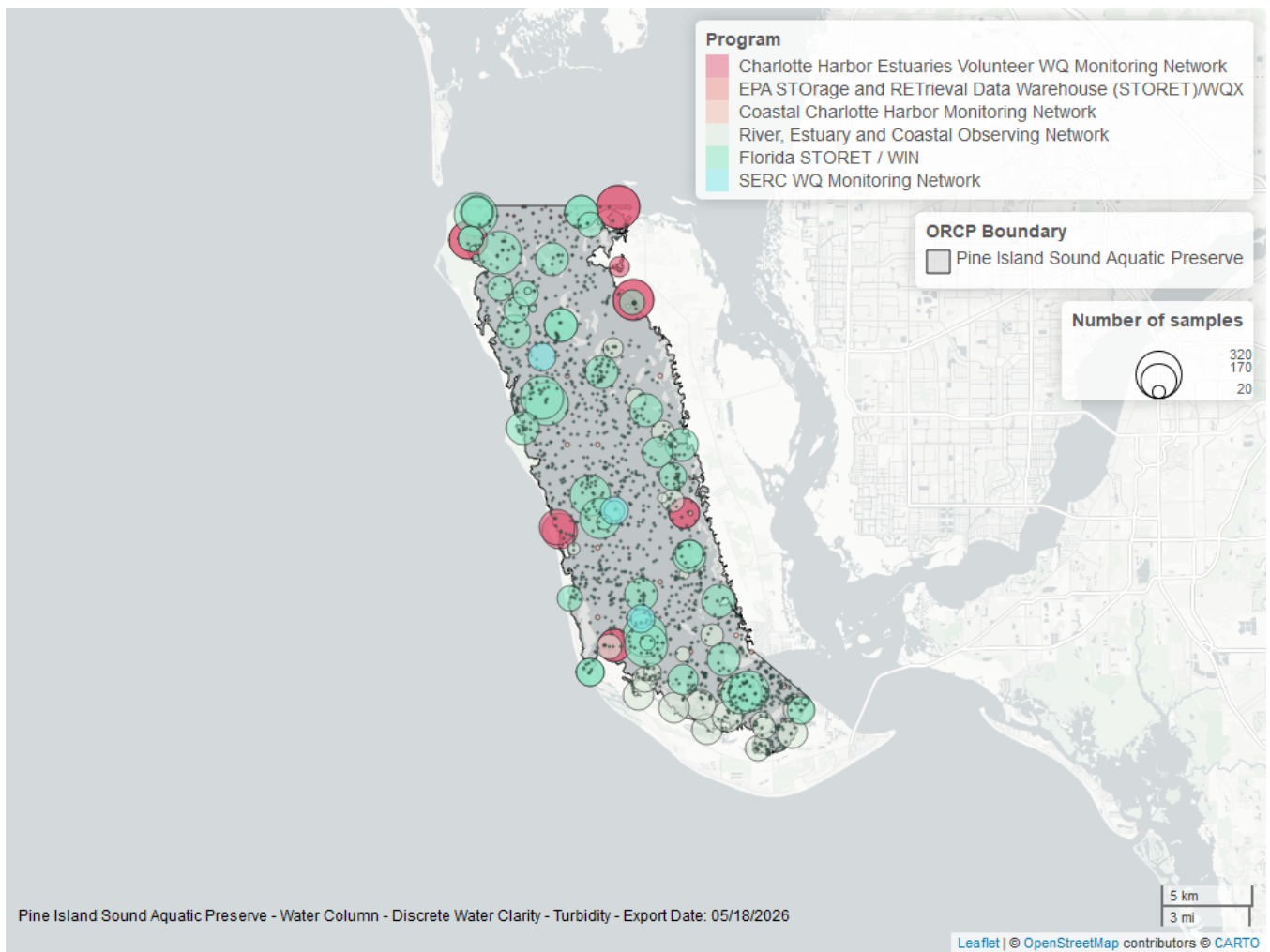


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	6935	1995	2025
303	2081	2012	2025
476	1351	1998	2025
509	348	1999	2008
513	169	2003	2025
103	2	2006	2006

Program names:

5002 - Florida STORET / WIN²

303 - River, Estuary and Coastal Observing Network¹

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³

509 - SERC Water Quality Monitoring Network⁵

513 - Coastal Charlotte Harbor Monitoring Network⁴

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

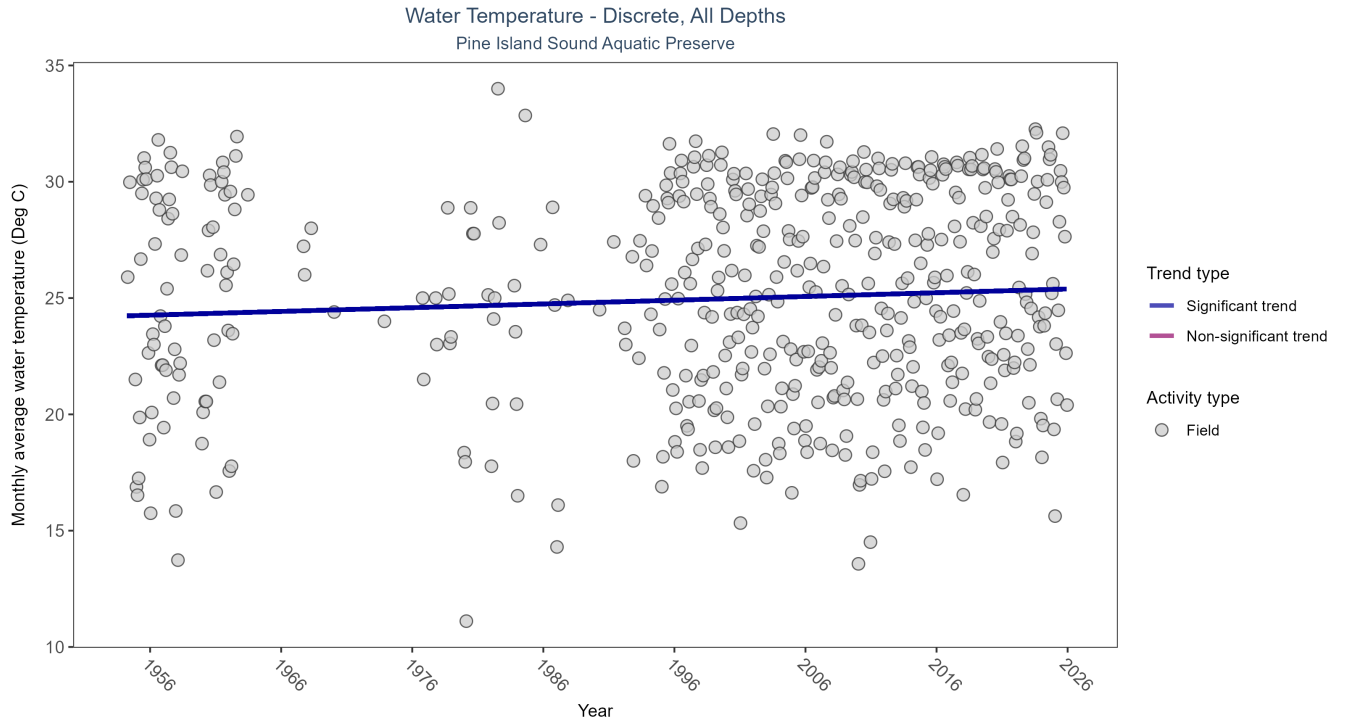


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	38470	62	1954 - 2025	26.4	0.1555	24.2323	0.0161	0

Monthly average water temperature increased by 0.02°C per year.

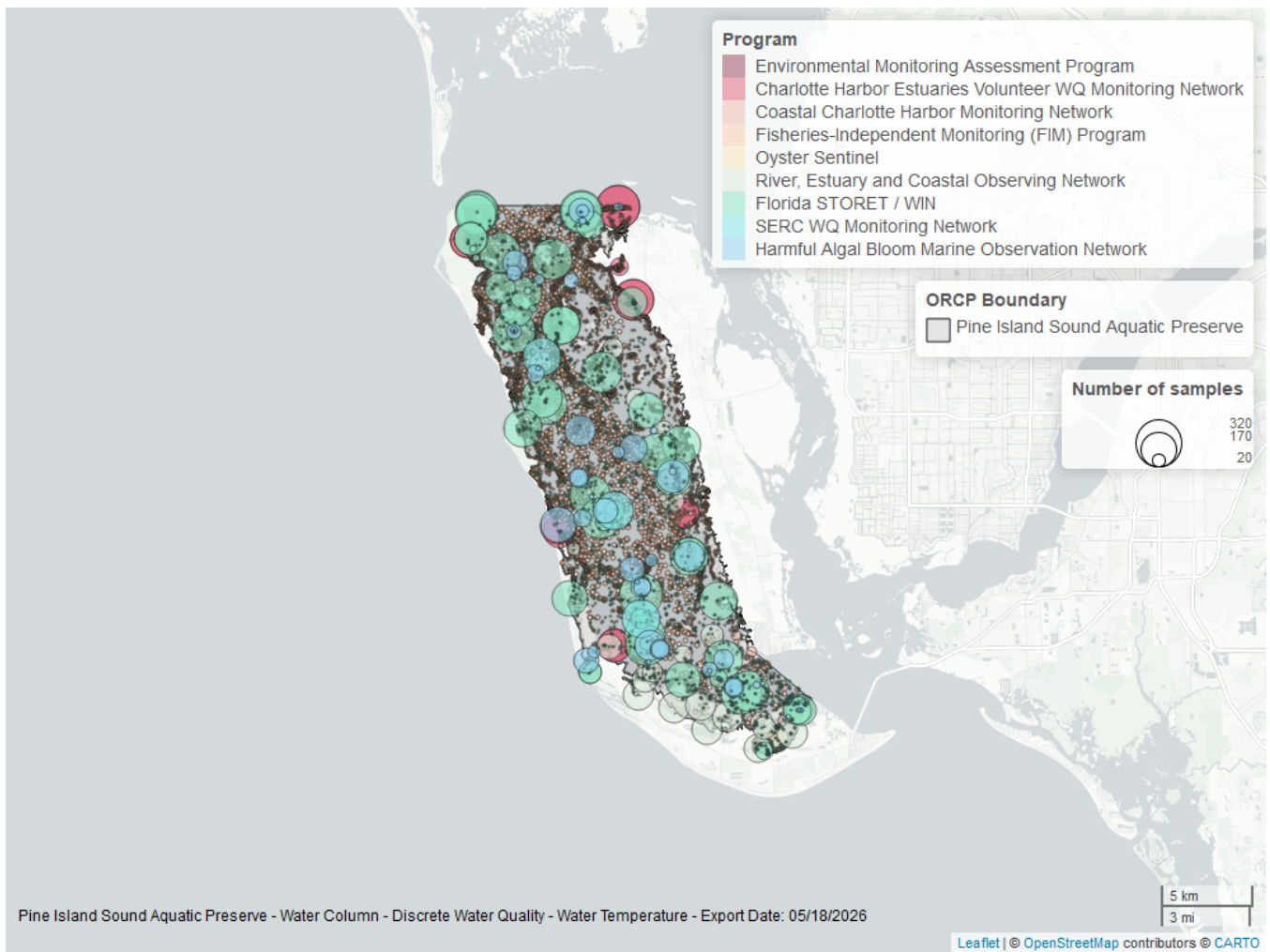


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
69	21551	1990	2024
5002	9873	1987	2025
95	2611	1954	2018
303	2158	2012	2025
476	1298	1998	2025
509	702	1999	2008
513	193	2002	2025
456	71	1970	2012
115	14	2002	2004

Program names:

69 - Fisheries-Independent Monitoring (FIM) Program¹⁰

5002 - Florida STORET / WIN²

95 - Harmful Algal Bloom Marine Observation Network¹¹

303 - River, Estuary and Coastal Observing Network¹
476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network³
509 - SERC Water Quality Monitoring Network⁵
513 - Coastal Charlotte Harbor Monitoring Network⁴
456 - Oyster Sentinel¹²
115 - Environmental Monitoring Assessment Program⁸

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Pine Island Sound Aquatic Preserve

Table 28: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
7	02293249	4	FALSE	pH , Dissolved Oxygen , Turbidity , Water Temperature, Salinity

Program names:

7 - National Water Information System¹³

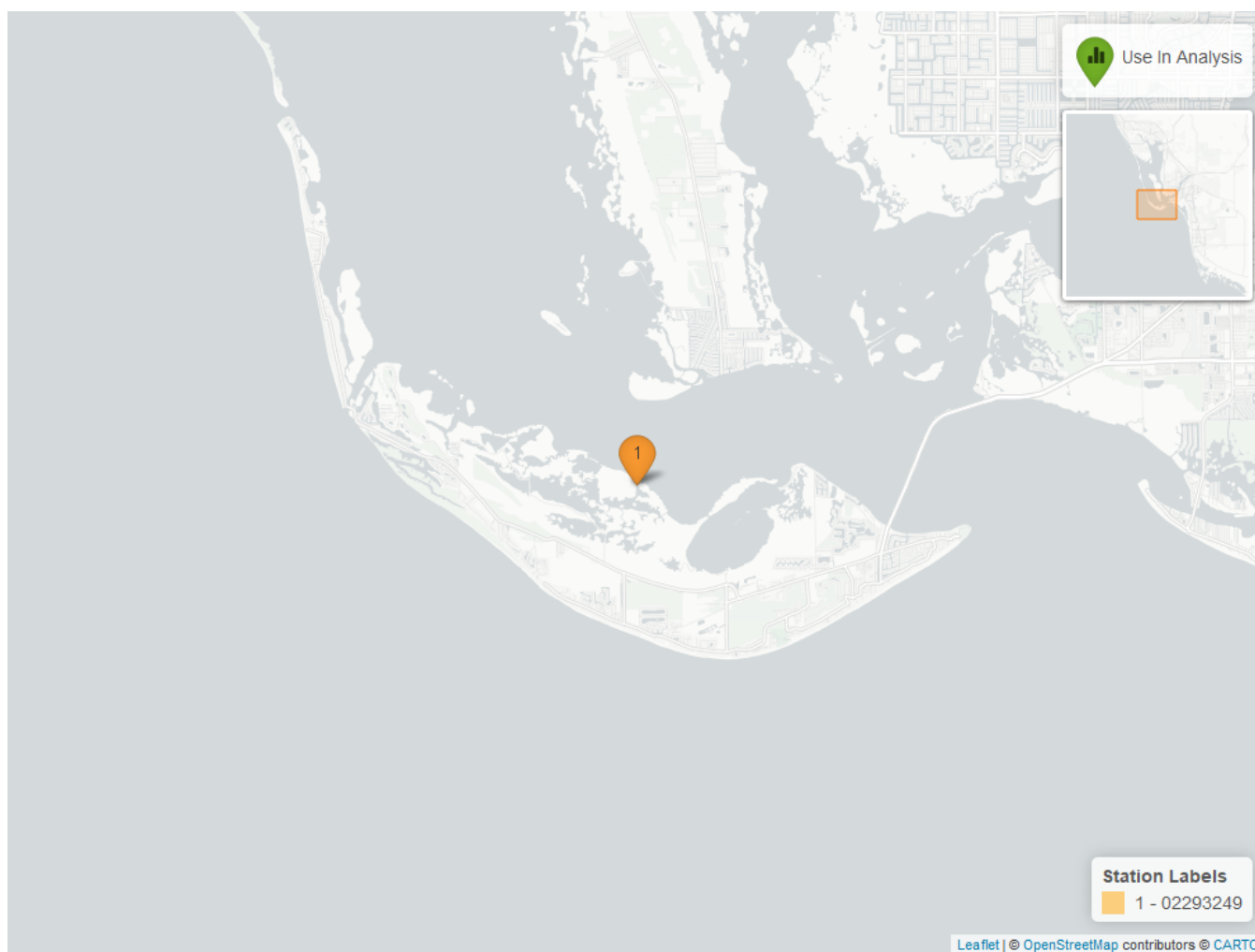


Figure 23: Map showing continuous water quality sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

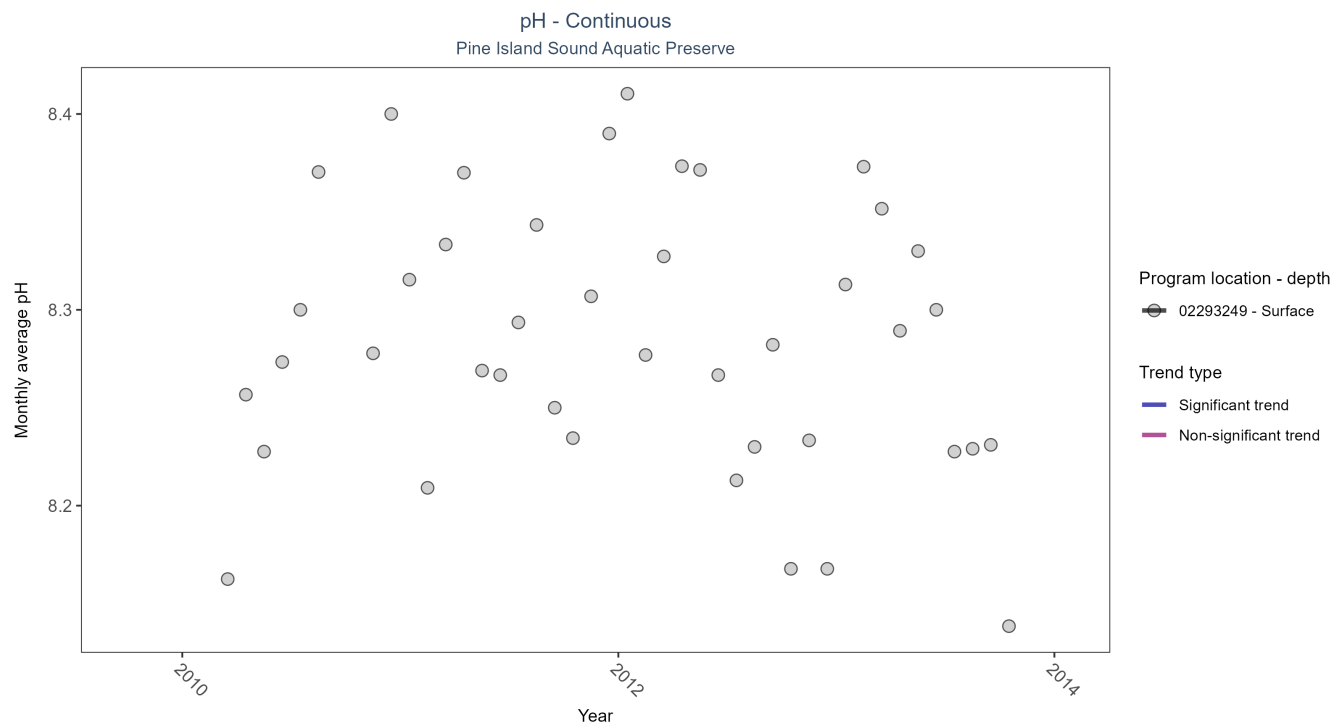


Figure 24: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02293249	Insufficient data to calculate trend	1164	4	2010 - 2013	8.3	-	-	-	-

There was insufficient data to fit a model for one location.

Dissolved Oxygen - Continuous

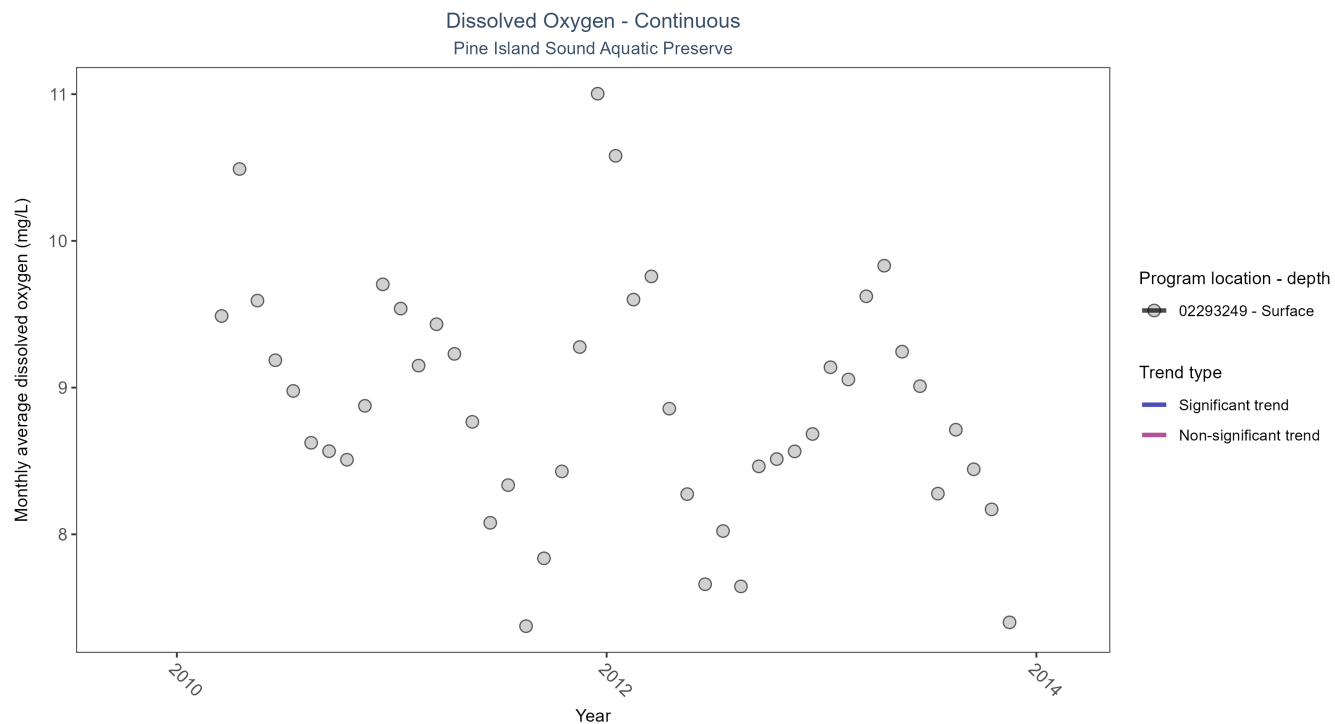


Figure 25: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02293249	Insufficient data to calculate trend	1302	4	2010 - 2013	8.8	-	-	-	-

There was insufficient data to fit a model for one location.

Turbidity - Continuous

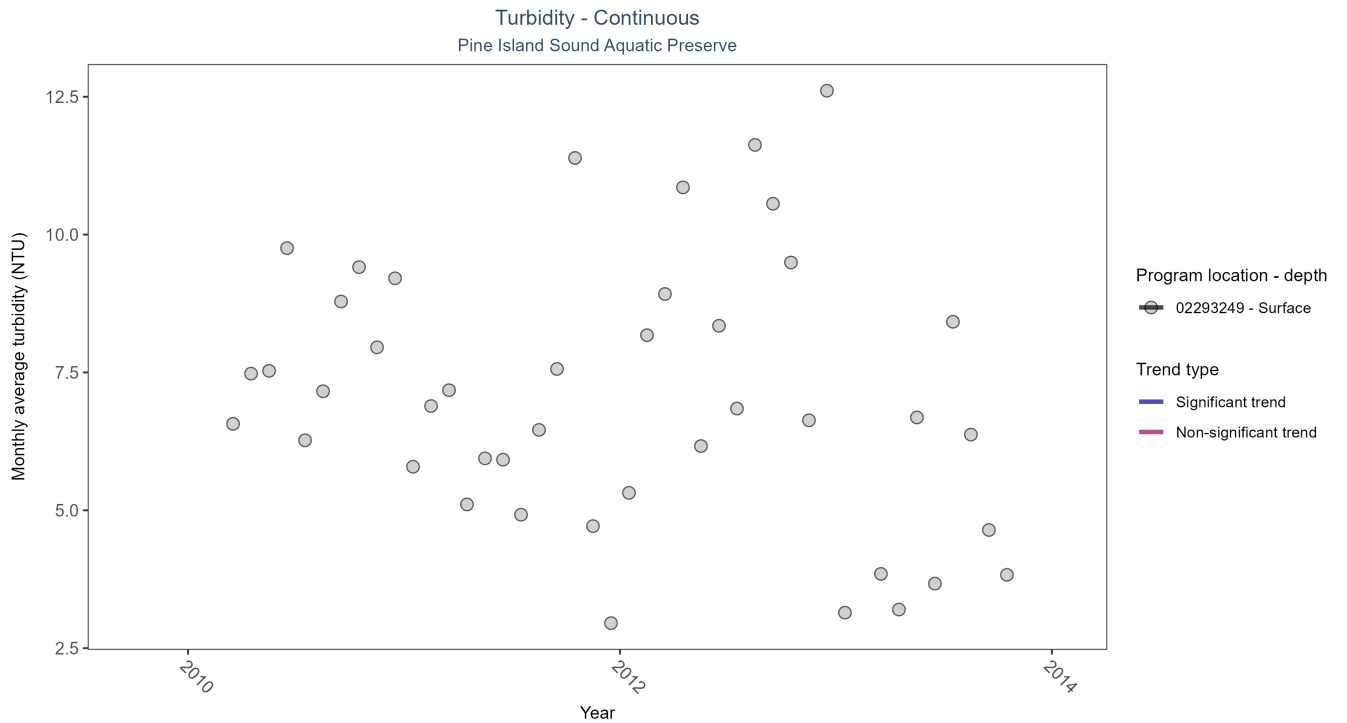


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02293249	Insufficient data to calculate trend	1174	4	2010 - 2013	5.4	-	-	-	-

There was insufficient data to fit a model for one location.

Water Temperature - Continuous

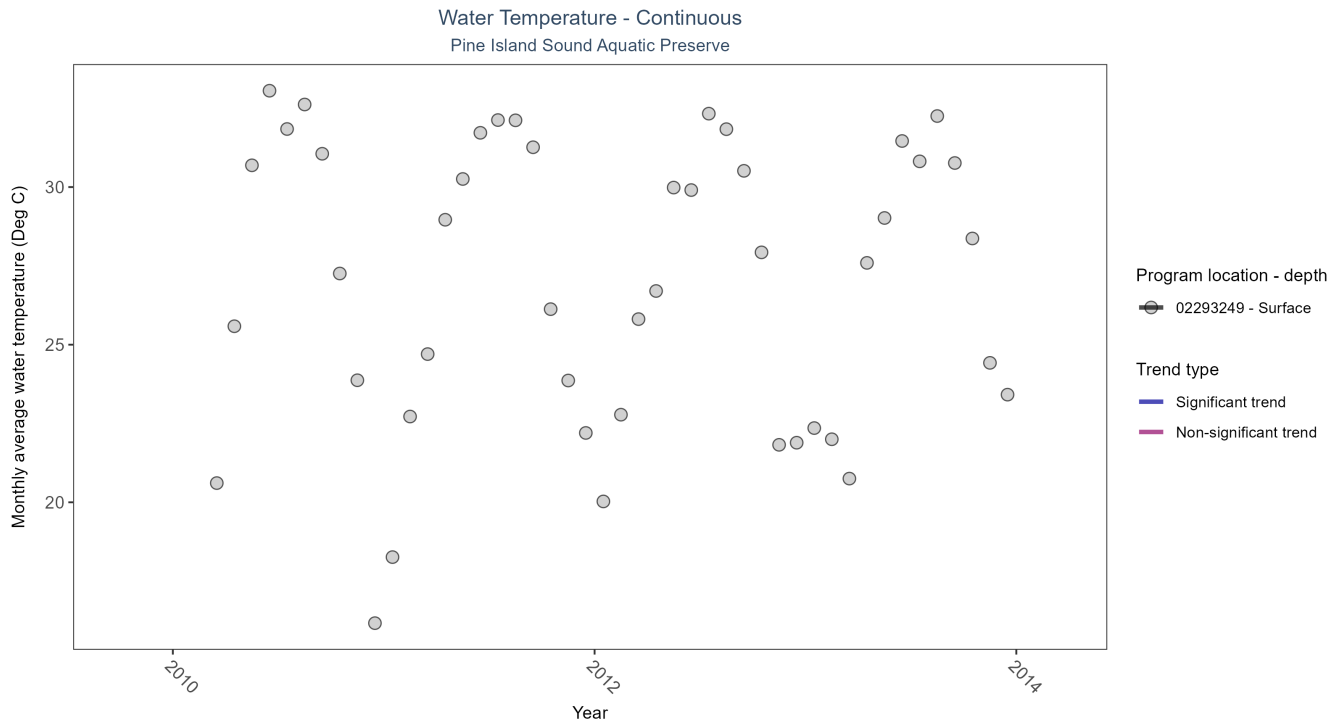


Figure 27: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02293249	Insufficient data to calculate trend	2445	4	2010 - 2013	28.1	-	-	-	-

There was insufficient data to fit a model for one location.

Salinity - Continuous

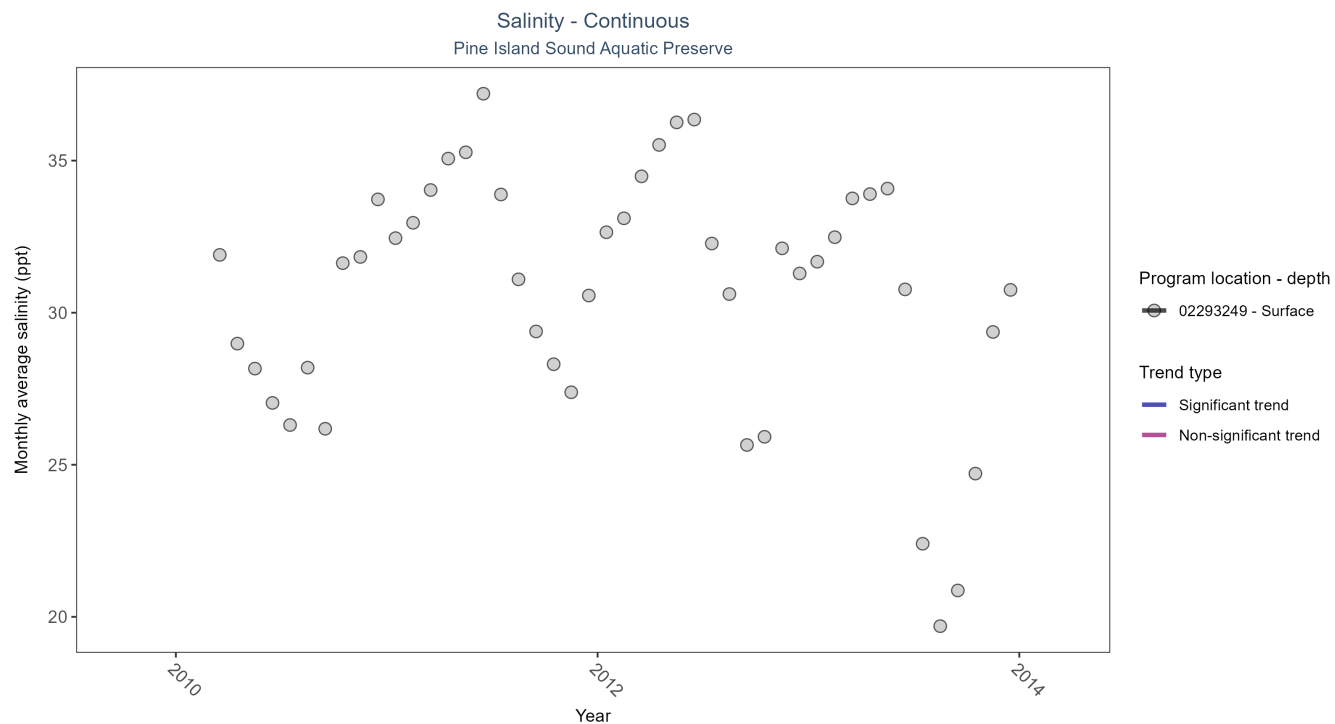


Figure 28: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02293249	Insufficient data to calculate trend	1871	4	2010 - 2013	32	-	-	-	-

There was insufficient data to fit a model for one location.

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila* spp. are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

Pine Island Sound Aquatic Preserve
SAV Percent Cover - Sample Locations

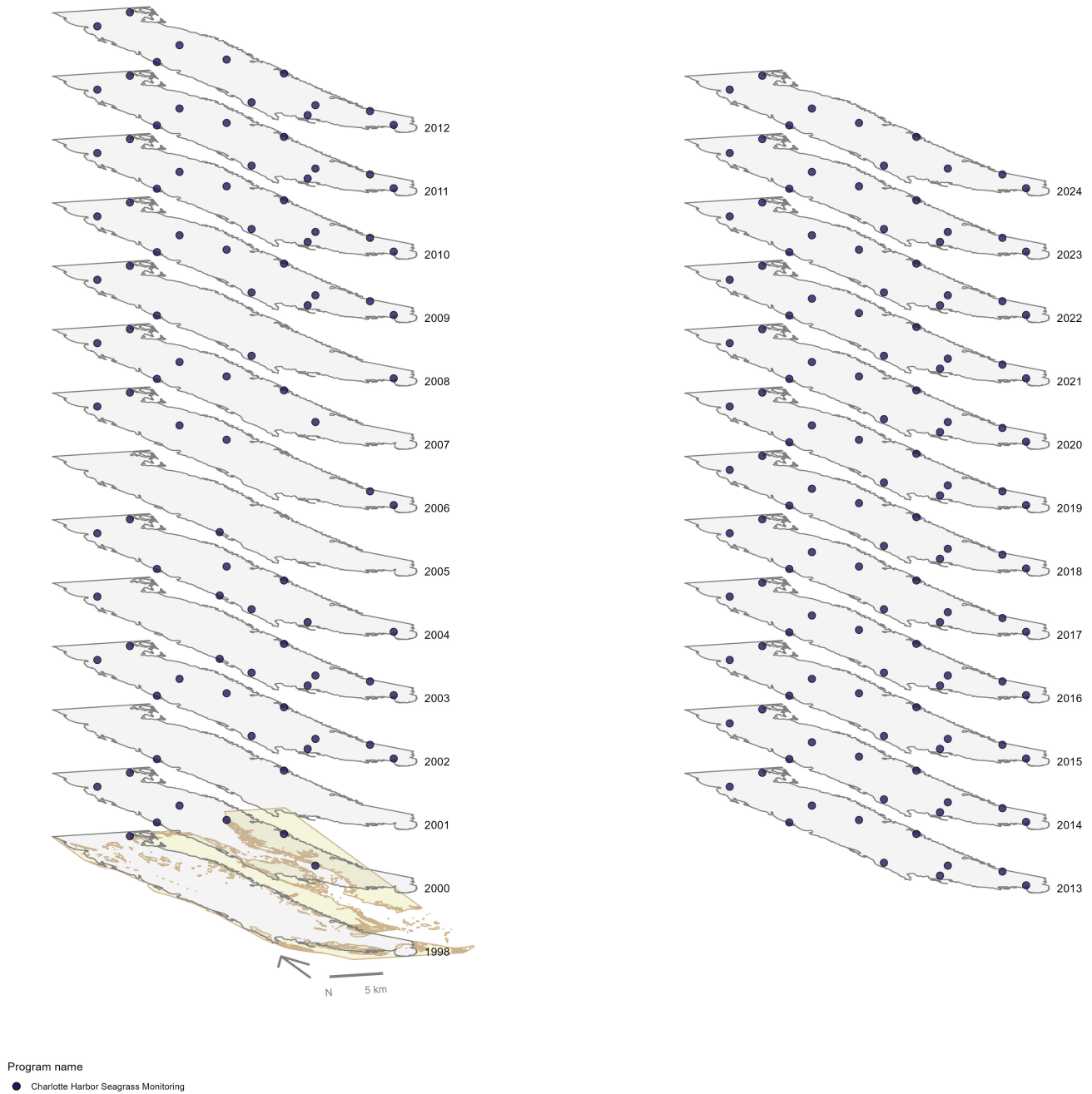


Figure 29: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Pine Island Sound Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

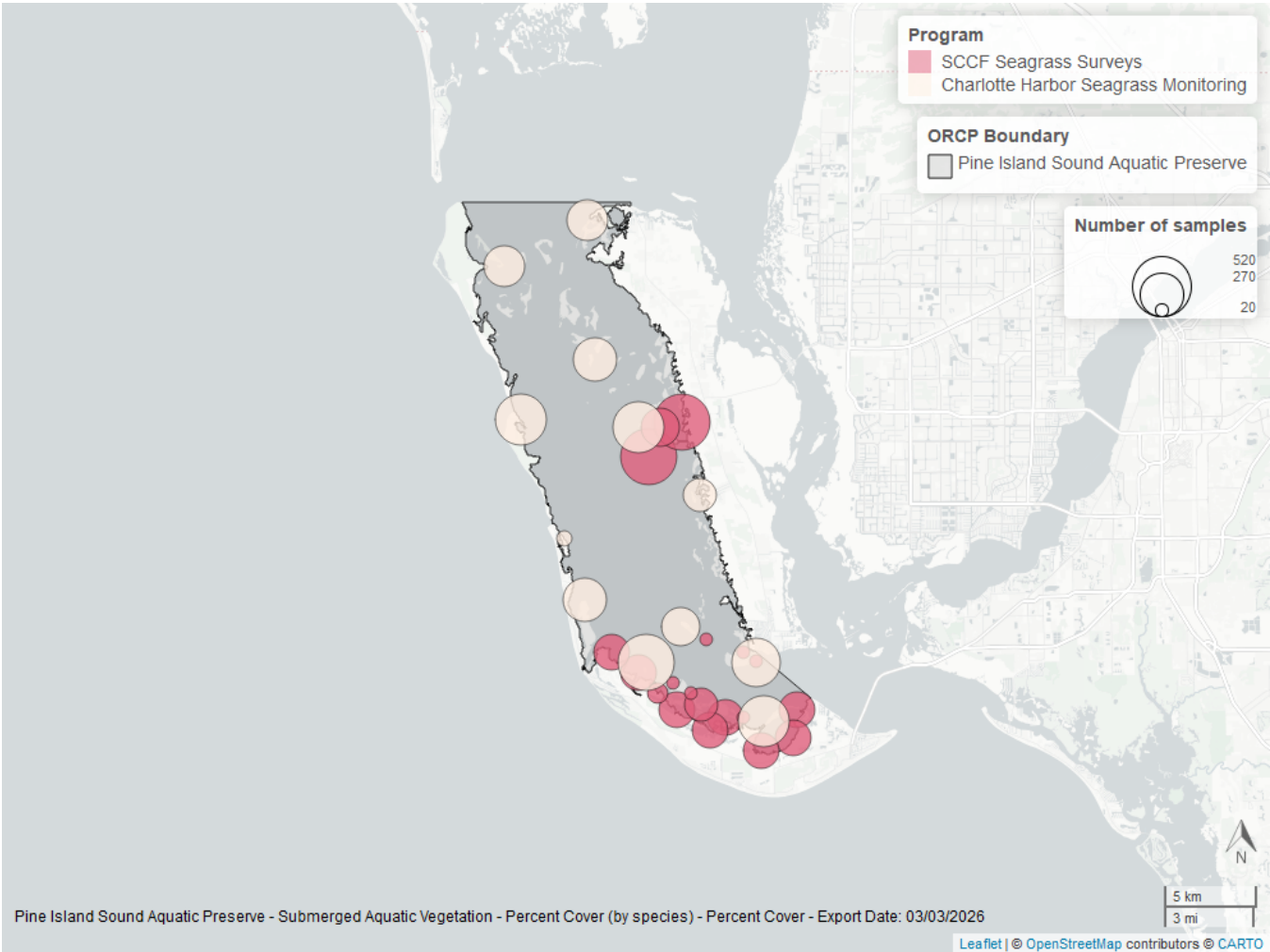


Figure 30: Map showing SAV sampling sites within the boundaries of *Pine Island Sound Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 34: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
570	3599	1998	2024	Braun Blanquet	12
3015	3179	2010	2024	Percent Occurrence	19

Program names:

570 - Charlotte Harbor Seagrass Monitoring¹⁴
3015 - SCCF Seagrass Surveys¹⁵

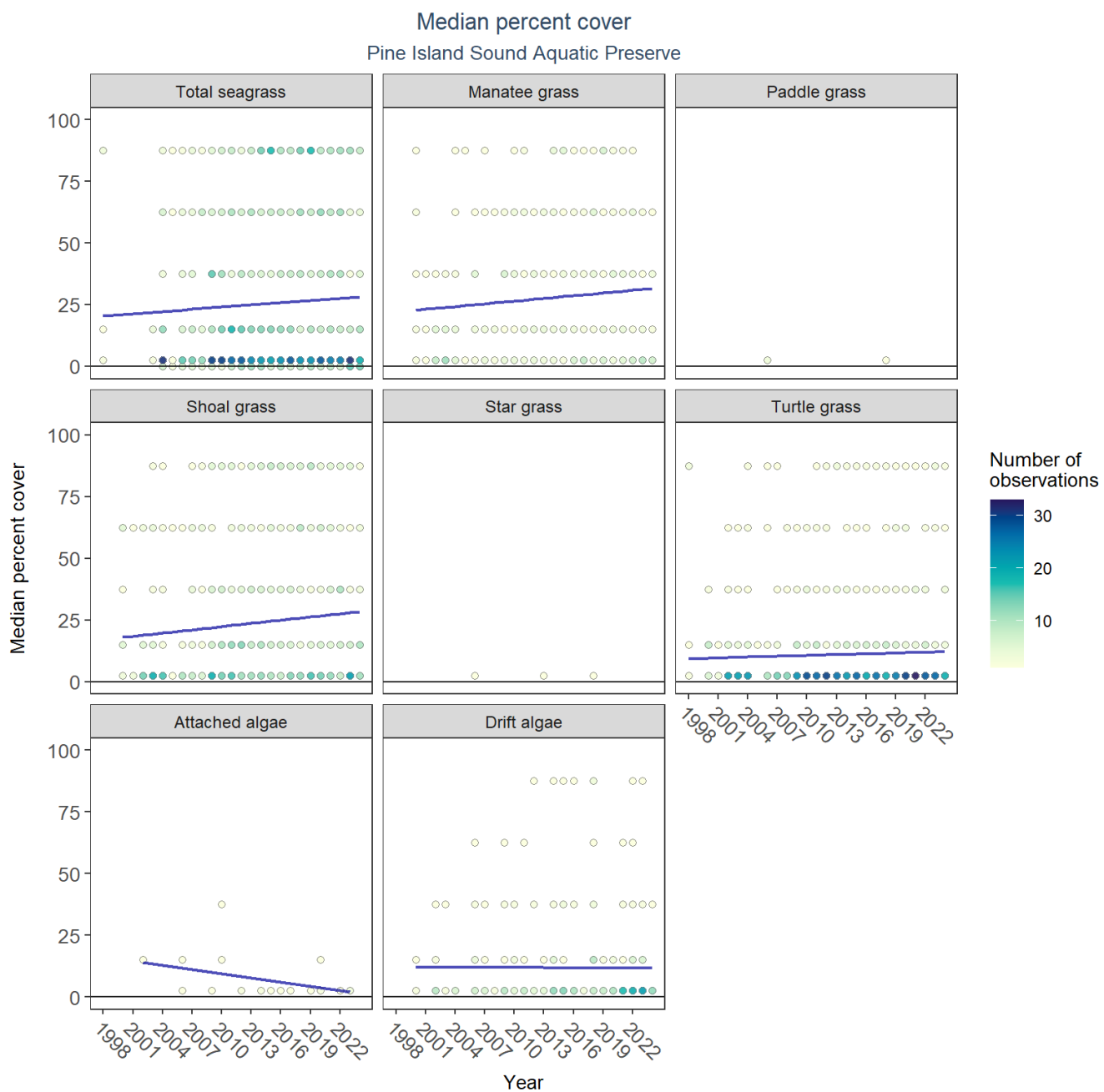


Figure 31: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

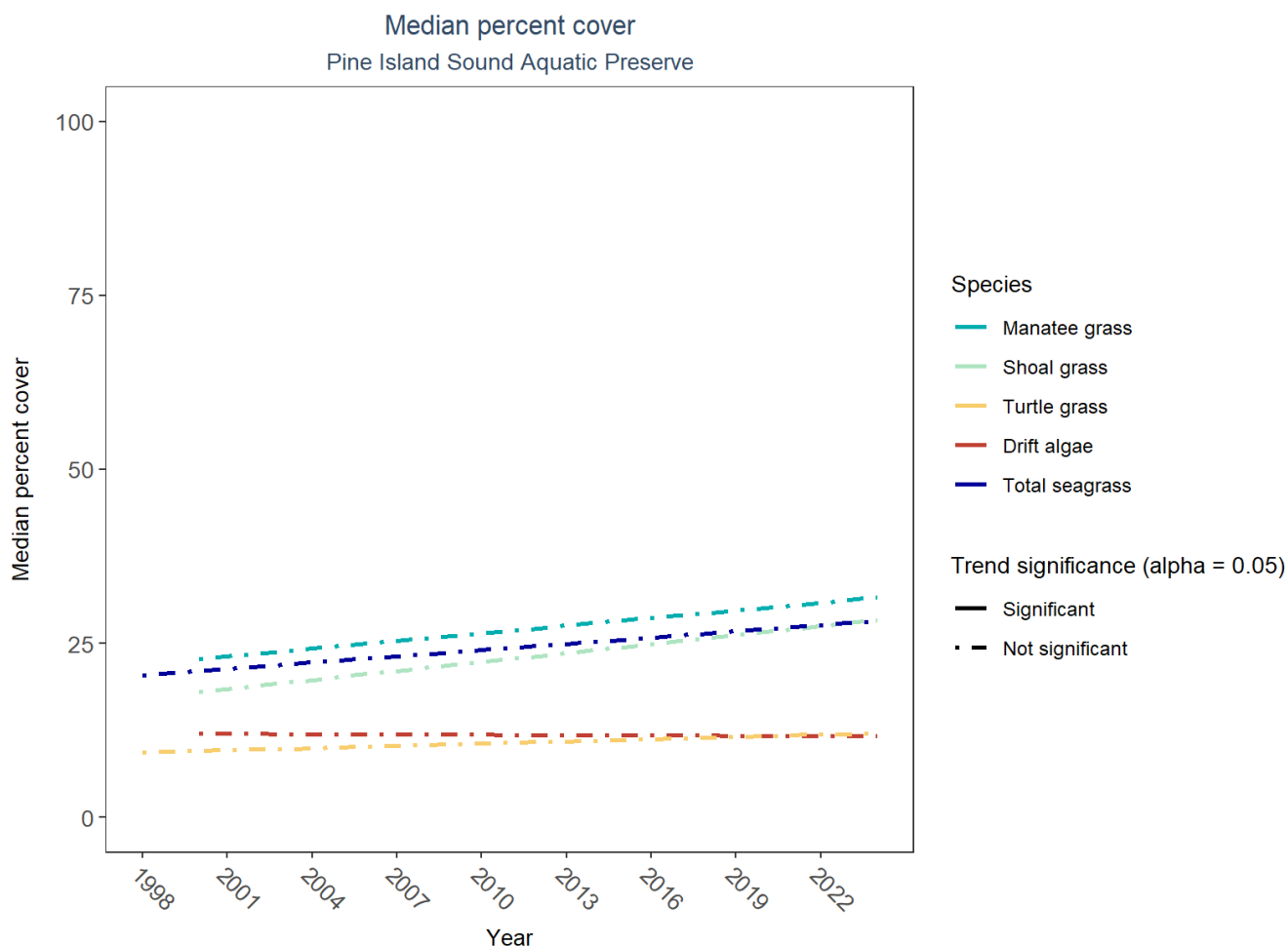


Figure 32: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 35: Percent Cover Trend Analysis for Pine Island Sound Aquatic Preserve

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Attached algae	No detectable trend	2002 - 2023	18.485846	-0.5710769	0.0671357
Drift algae	No detectable trend	2000 - 2024	12.125756	-0.0149842	0.9318566
Shoal grass	No detectable trend	2000 - 2024	15.443261	0.4301651	0.2192753
Paddle grass	Insufficient data	-	-	-	-
Star grass	Insufficient data	-	-	-	-
Manatee grass	No detectable trend	2000 - 2024	20.588525	0.3662709	0.2832966
Turtle grass	No detectable trend	1998 - 2024	8.907789	0.1063973	0.4595171
Total seagrass	No detectable trend	1998 - 2024	19.273743	0.2983261	0.3284210

Total seagrass, manatee grass, shoal grass, turtle grass, attached algae, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for paddle grass and star grass due to insufficient data.

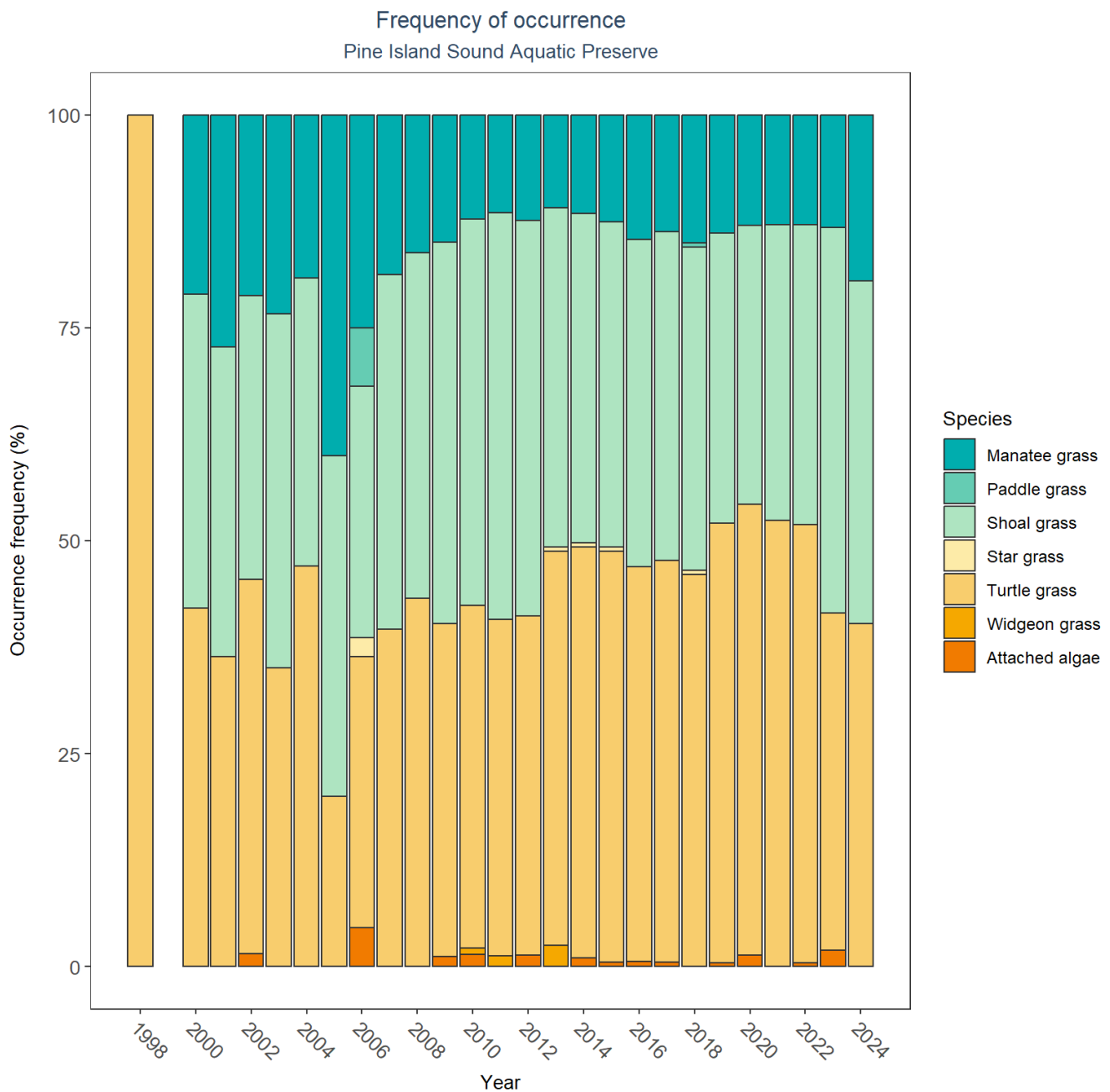


Figure 33: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Pine Island Sound Aquatic Preserve within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity

- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Coastal Wetlands

The data file used is: **All_CW_Parameters-2026-Jun-04.txt**

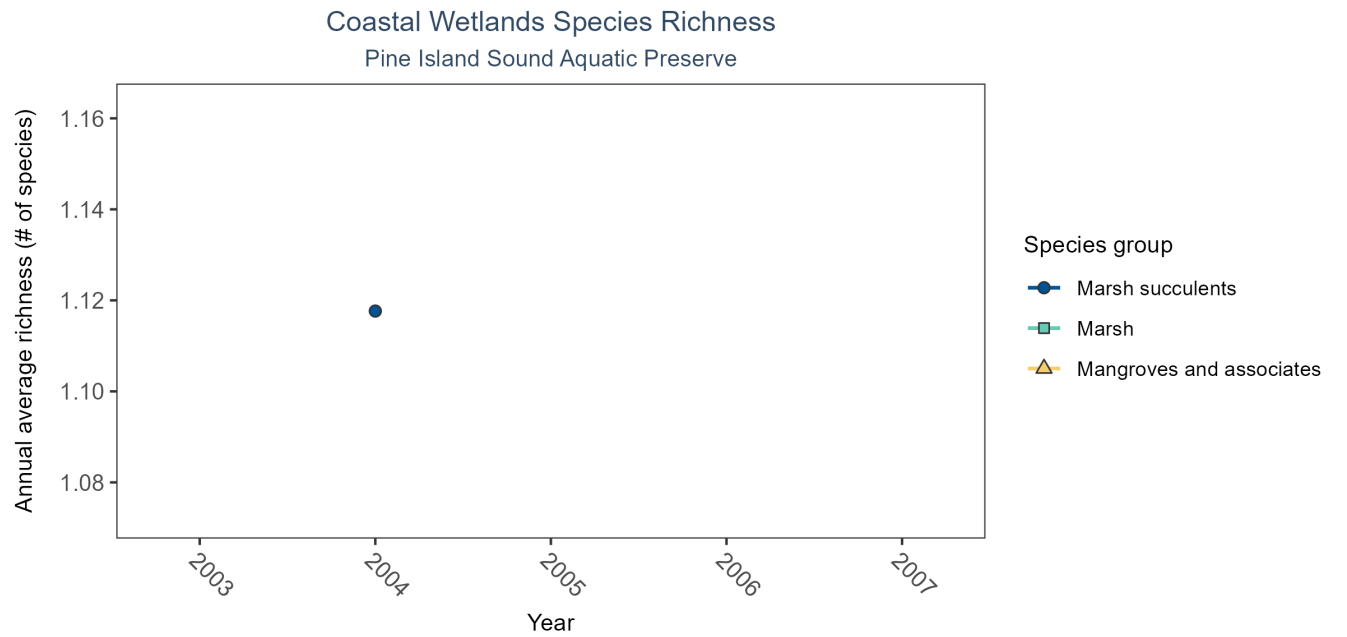


Figure 34: Line graph of annual average coastal wetlands species richness over time for mangroves and associates (triangles), marsh (squares), and marsh succulents (circles). If the time series by species group included more than one year of observations, a line connects data points for visualization.

Table 36: Coastal Wetlands Species Richness

<i>Species Group</i>	<i>No. of Samples</i>	<i>No. Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
Marsh succulents	17	1	2004 - 2004	1	1.12

In the year 2004, 1 species were observed for marsh succulents based on 17 observations.

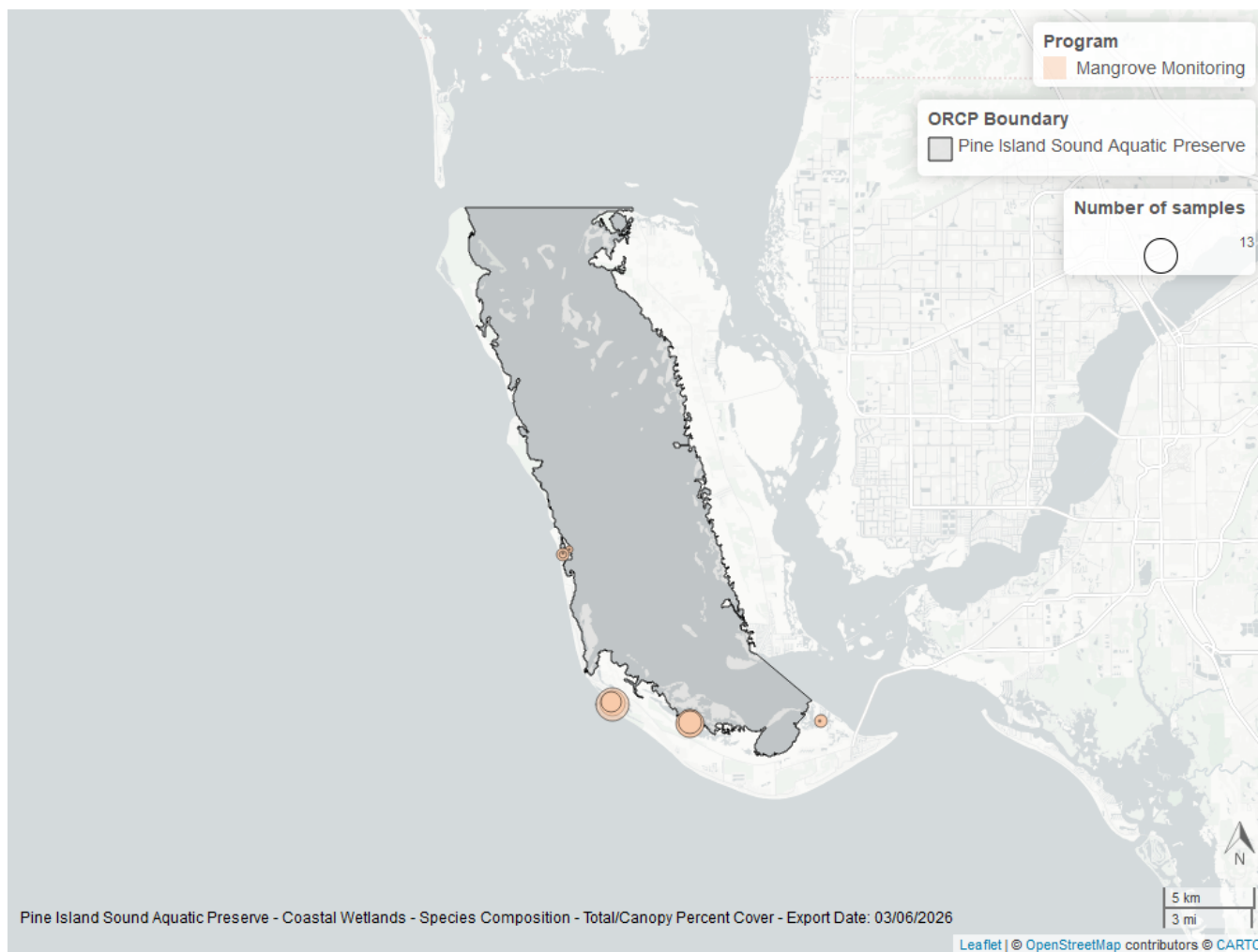


Figure 35: Map showing location of coastal wetlands sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Oyster

The data file used is: **All_OYSTER_Parameters-2026-Jun-04.txt**

Pine Island Sound Aquatic Preserve

Oyster sample locations available in SEACAR by Program and Year

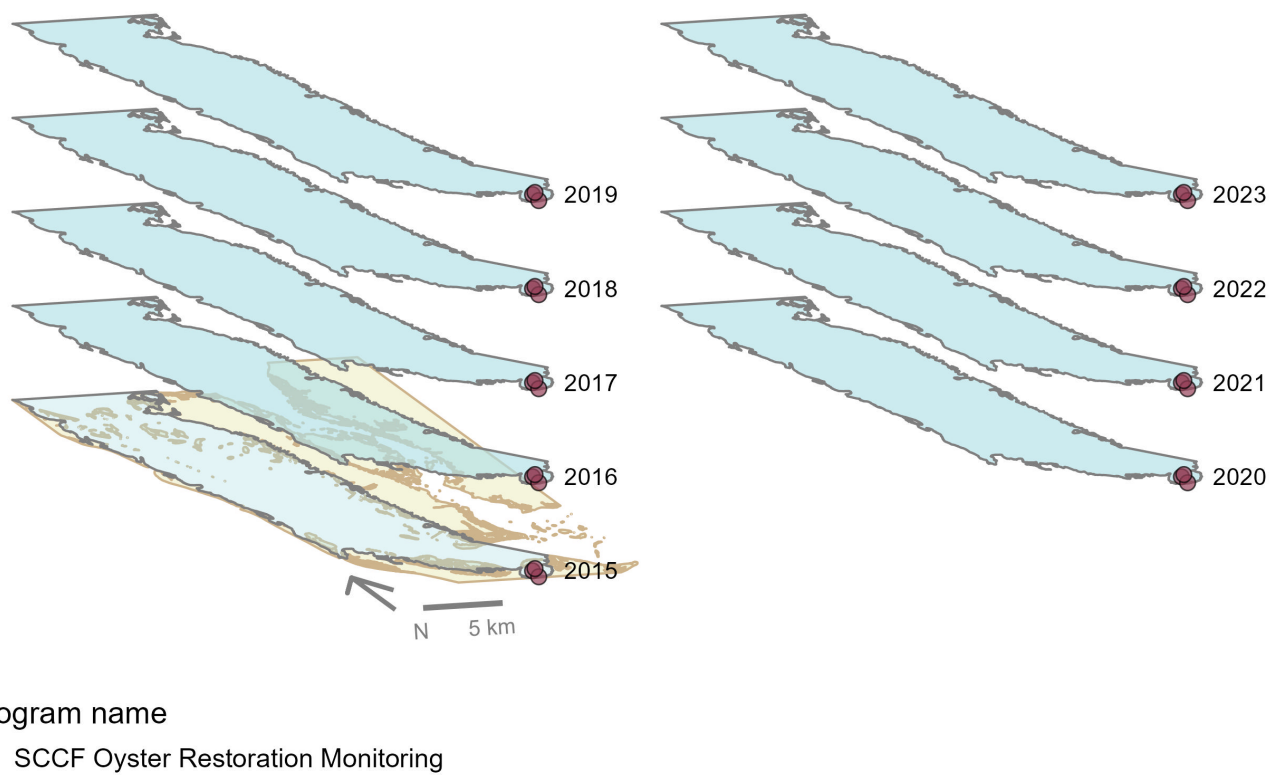


Figure 36: Maps showing the temporal scope of oyster sampling sites within the boundaries of *Pine Island Sound Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Density

For natural reefs, density showed no detectable trend between 2015 and 2023. For restored reefs, density decreased by an average of 232.83 oysters per square meter per year.

Natural

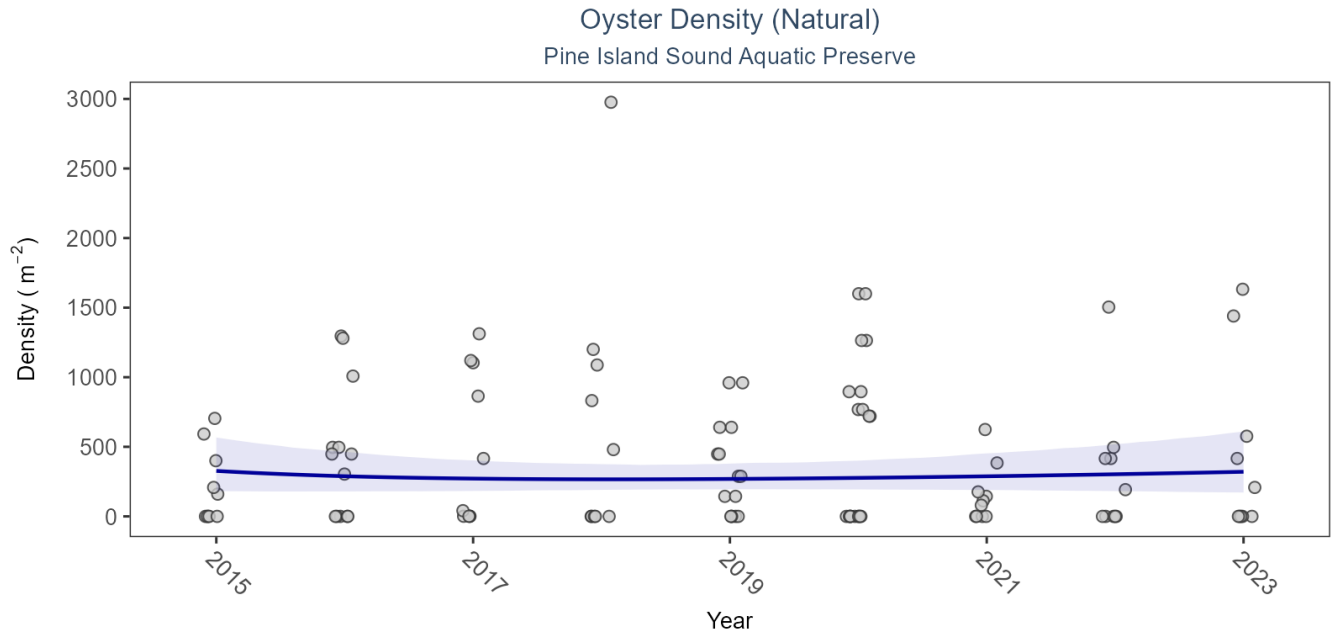


Figure 37: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 37: Model results for Oyster Density - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	No detectable trend	2015 - 2023	-0.78	22.66	-43.54 to 46.91

Restored

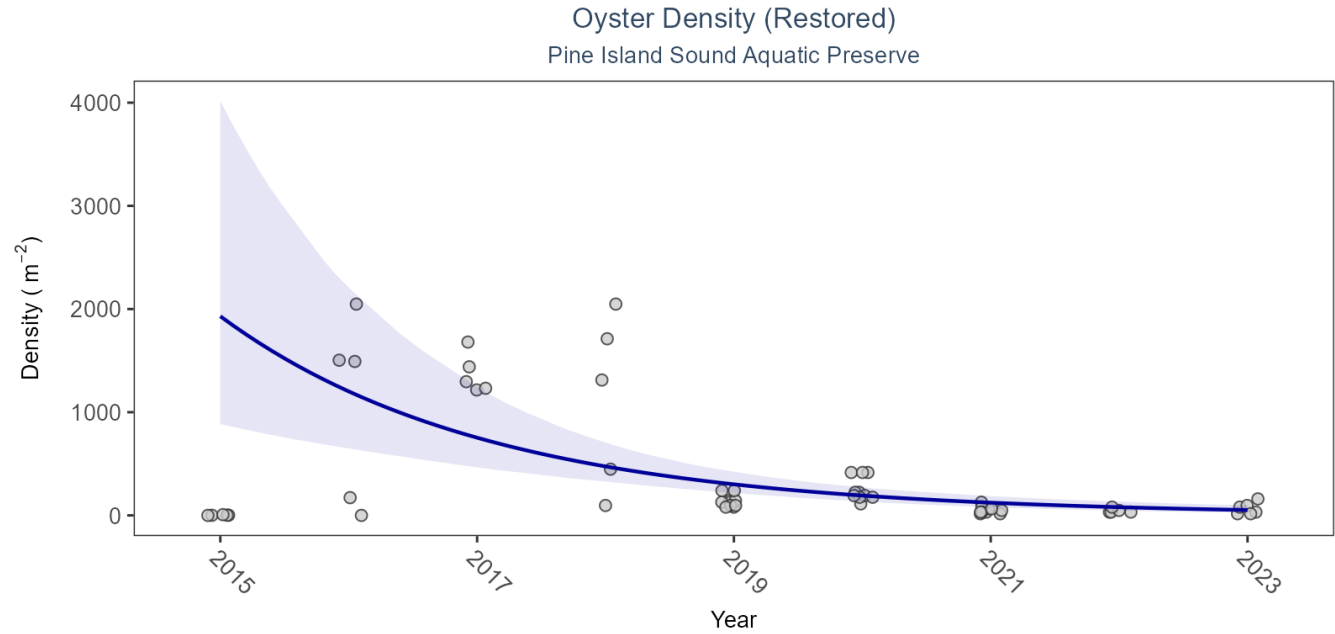


Figure 38: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 38: Model results for Oyster Density - Restored

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Restored	Live Oysters	Decreasing trend	2015 - 2023	-234.76	98.36	-499.45 to -100.94

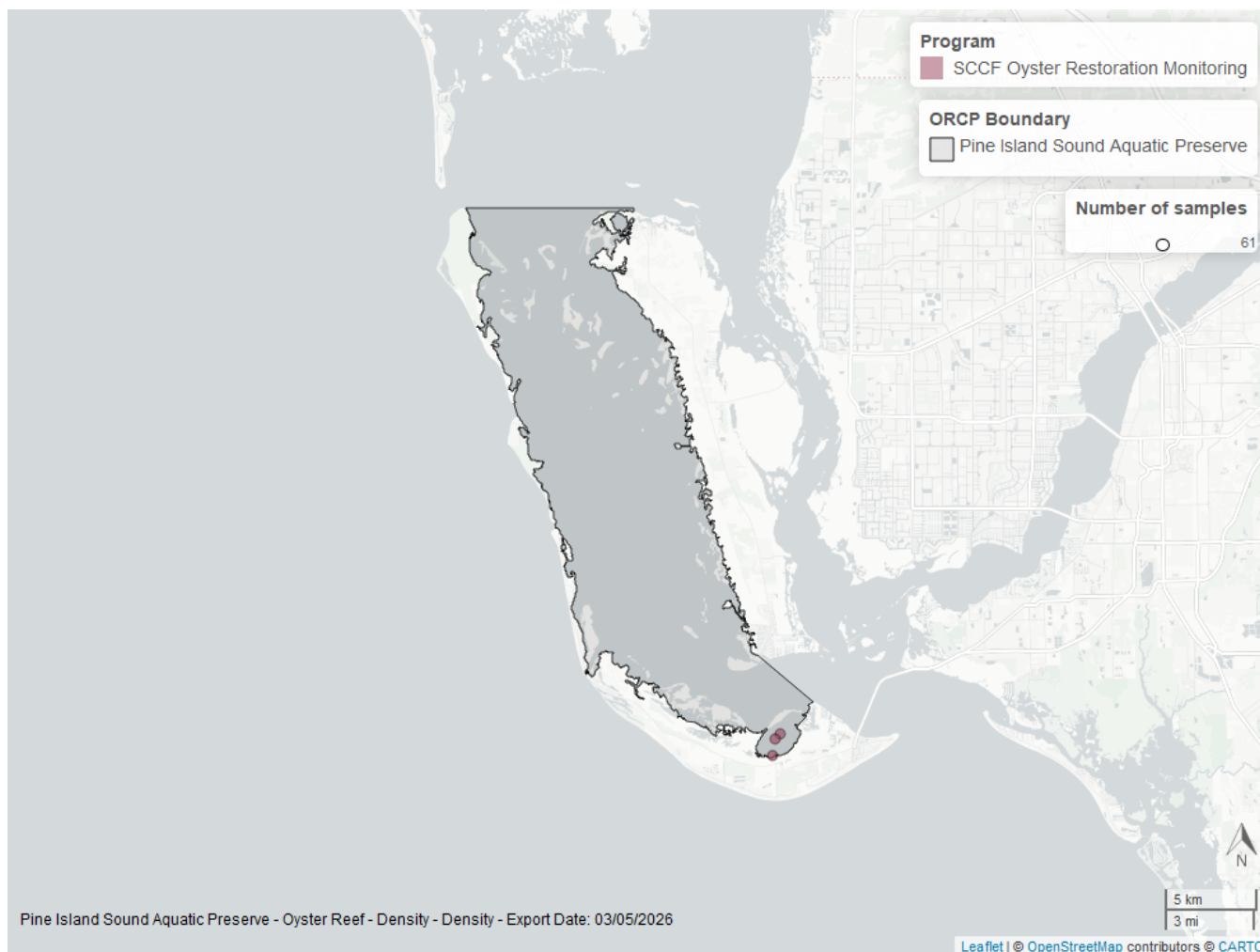


Figure 39: Map showing location of oyster density sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Percent Live

For natural reefs, percent live cover increased by an average of 8.42% per year. For restored reefs, percent live cover showed no detectable trend between 2019 and 2023.

Natural

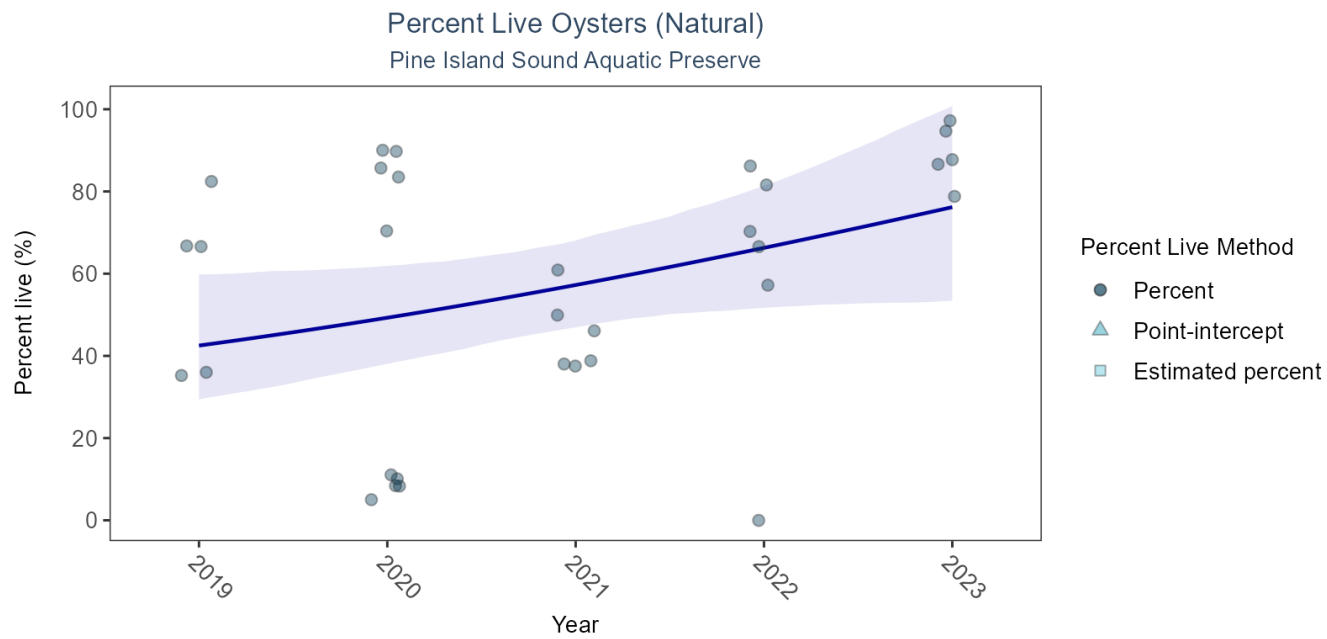


Figure 40: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 39: Model results for Oyster Percent Live - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	Increasing trend	2019 - 2023	8.42	4.01	0.62 to 16.49

Restored

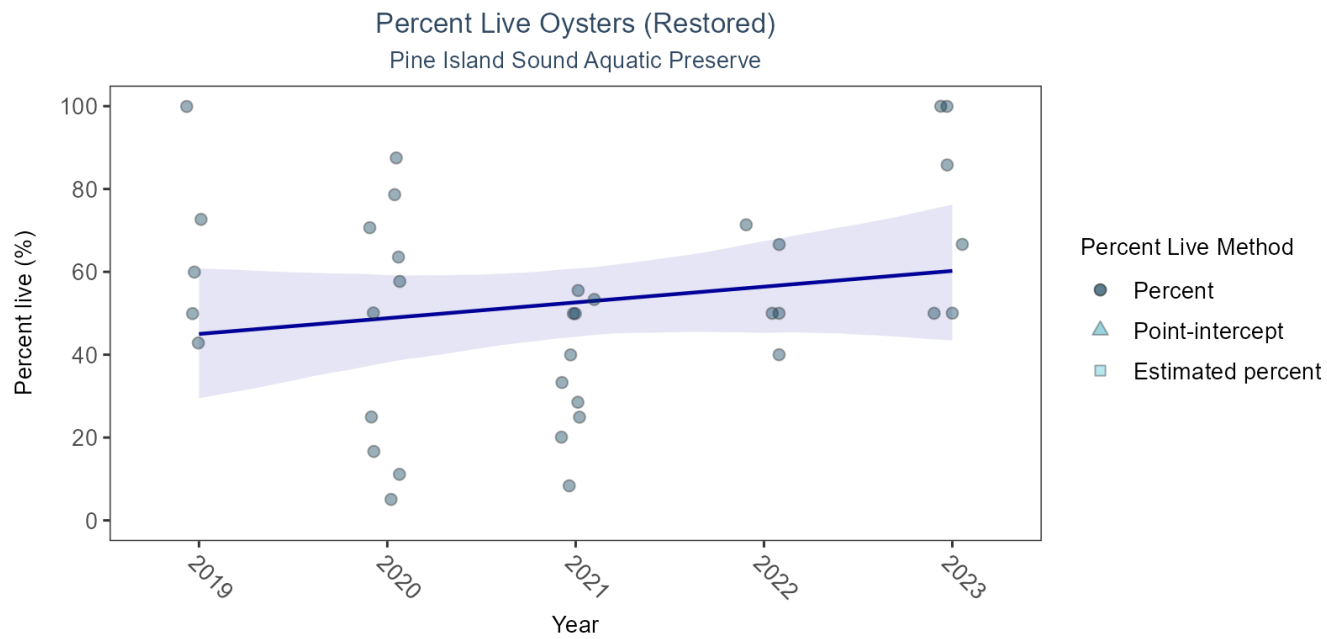


Figure 41: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 40: Model results for Oyster Percent Live - Restored

Habitat Type	Shell Type	Statistical Trend	Period of Record	Estimate	Standard Error	Credible Interval
Restored	Live Oysters	No detectable trend	2019 - 2023	3.8	3.44	-3.08 to 10.54

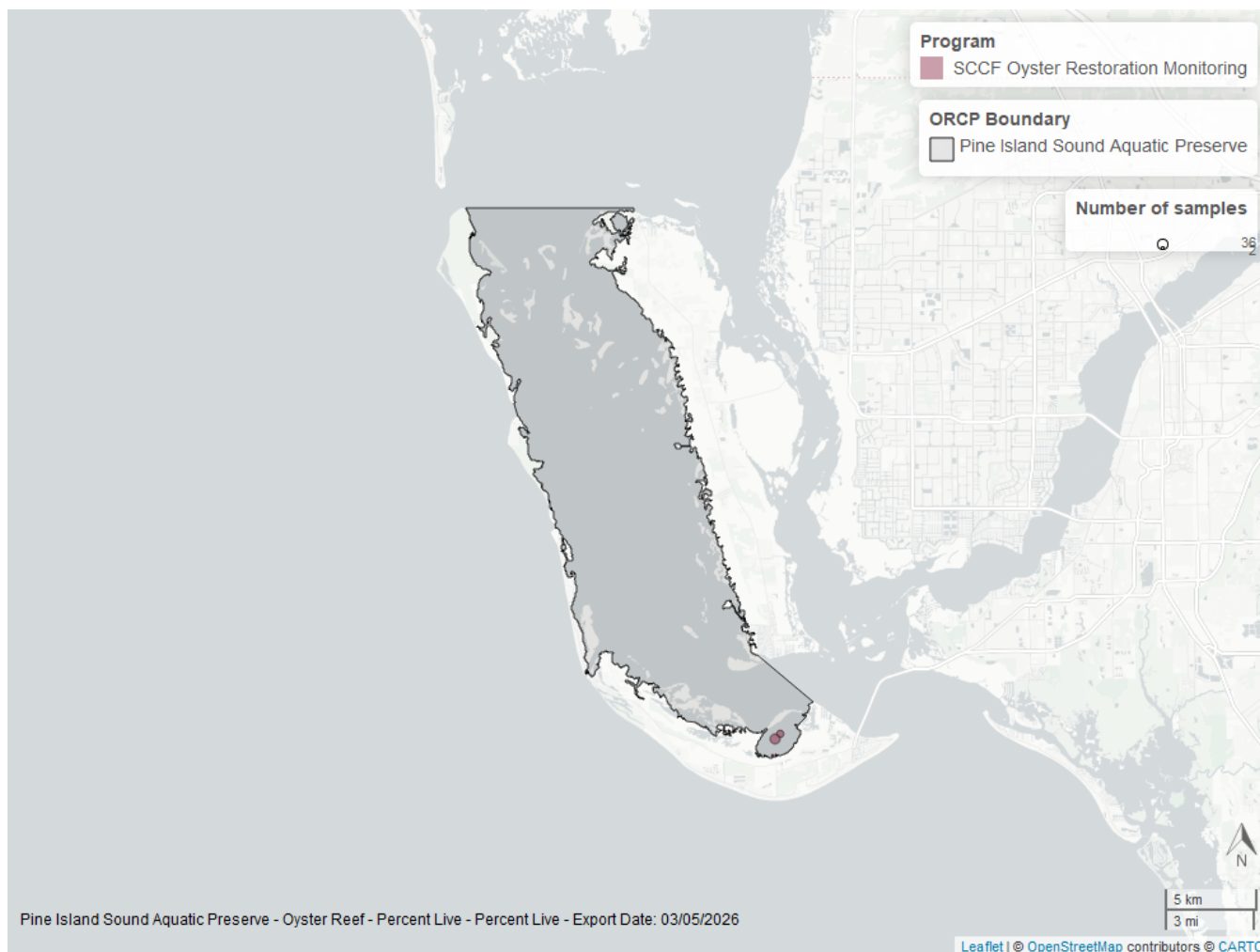


Figure 42: Map showing location of oyster percent live sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Shell Height

For natural reefs, annual average live oyster shell height in the 25-75mm size class decreased by 0.87mm per year, and there was no detectable trend for live oysters in the ≥ 75 mm size class. For restored reefs, annual average live oyster shell height in the ≥ 75 mm size class increased by 6.06mm per year, and there was no detectable trend for live oysters in the 25-75mm size class.

Natural

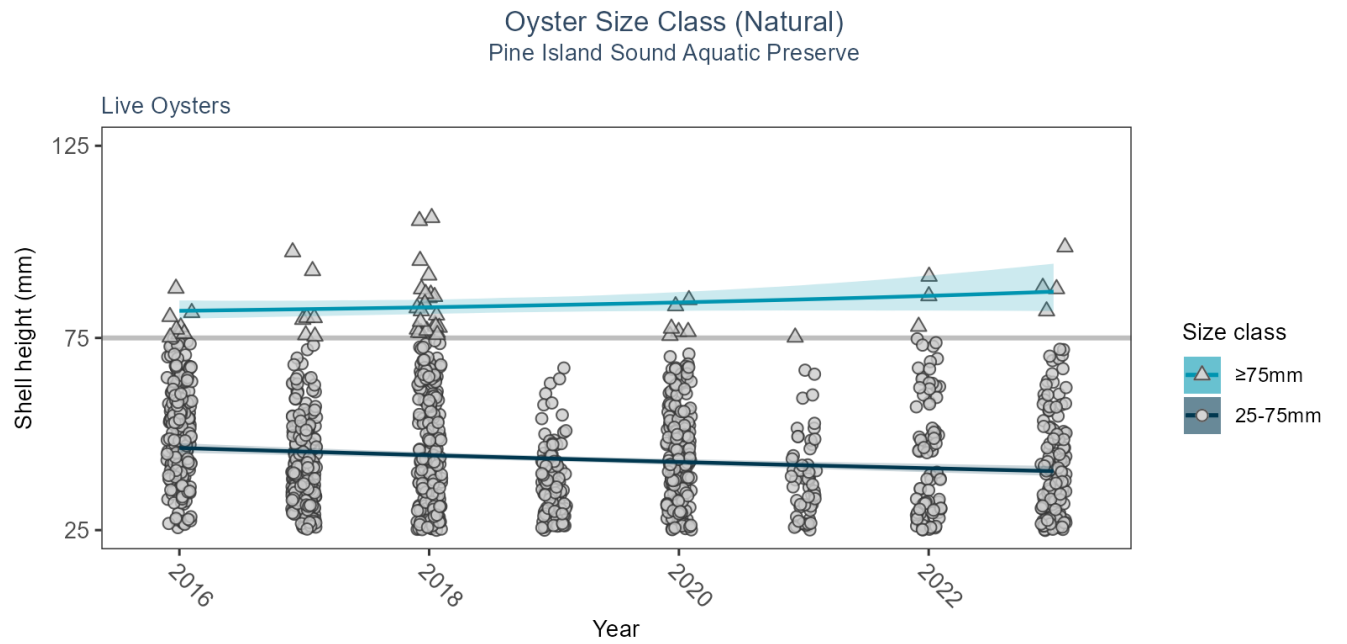


Table 41: Model results for Oyster Shell Height - Natural

Habitat Type	Shell Type	SizeClass	Statistical Trend	Period of Record	No. of Samples	Estimate	Standard Error	Credible Interval
Natural	Live Oysters	>75mm	No detectable trend	2016 - 2023	53	0.71	0.55	-0.24 to 1.87
Natural	Live Oysters	25-75mm	Decreasing trend	2016 - 2023	1224	-0.86	0.15	-1.16 to -0.55

Restored

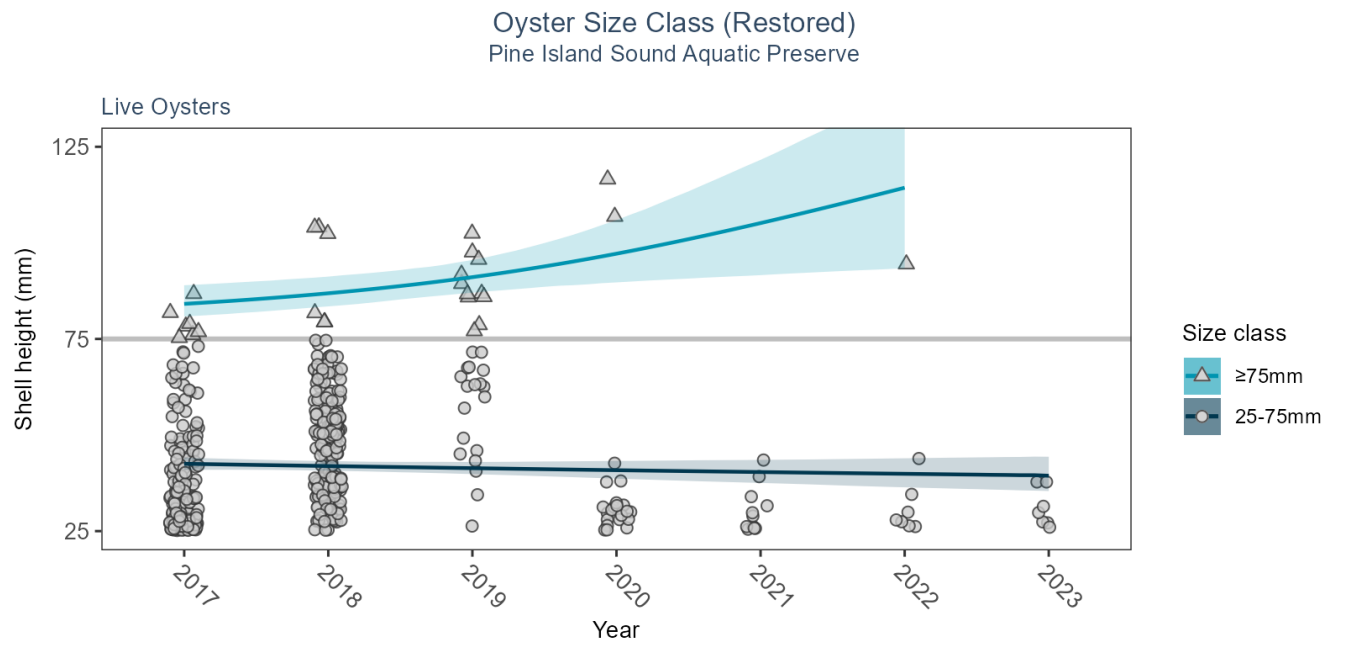


Table 42: Model results for Oyster Shell Height - Restored

<i>Habitat Type</i>	<i>Shell Type</i>	<i>SizeClass</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>No. of Samples</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Restored	Live Oysters	>75mm	Increasing trend	2017 - 2022	27	6.04	2.54	1.36 to 11.23
Restored	Live Oysters	25-75mm	No detectable trend	2017 - 2023	431	-0.52	0.69	-1.32 to 0.42

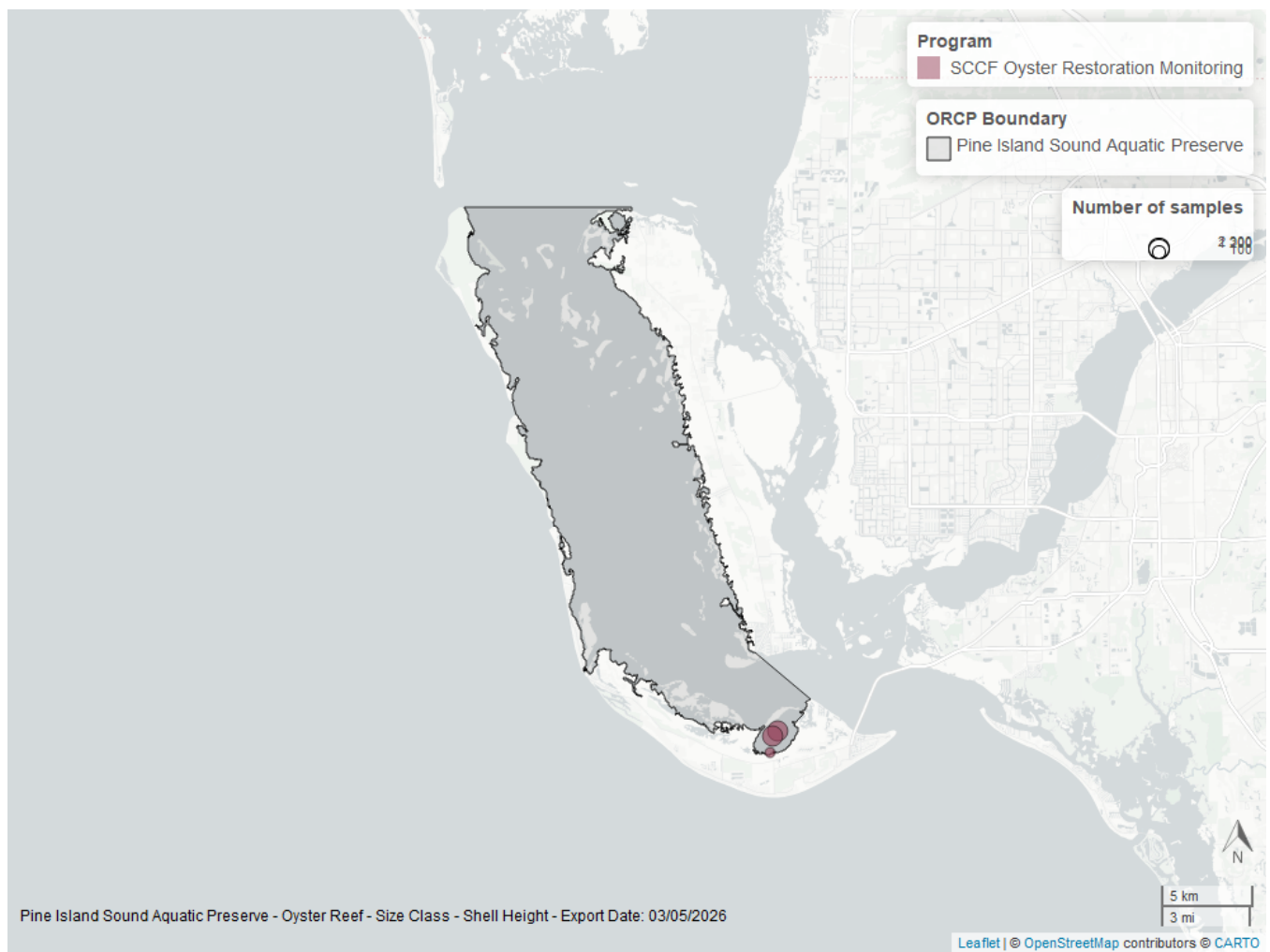


Figure 43: Map showing location of oyster shell height sampling locations within the boundaries of *Pine Island Sound Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Batis maritima ²	Halodule wrightii ¹	Salicornia virginica ²
Borrchia spp.	Halophila decipiens ¹	Sesuvium spp. ²
Caulerpa prolifera ¹	Halophila engelmannii ¹	Syringodium filiforme ¹
Caulerpa sertularioides ¹	Macroalgae ¹	Thalassia testudinum ¹
Caulerpa spp. ¹	No grass in quadrat ¹	Total seagrass ¹
Drift algae ¹	Ruppia maritima ¹	Unidentified species

1 - Submerged Aquatic Vegetation, 2 - Coastal Wetlands

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